

Axilla

Department of Anatomy
Musculoskeletal System
The Upper Limb (3)

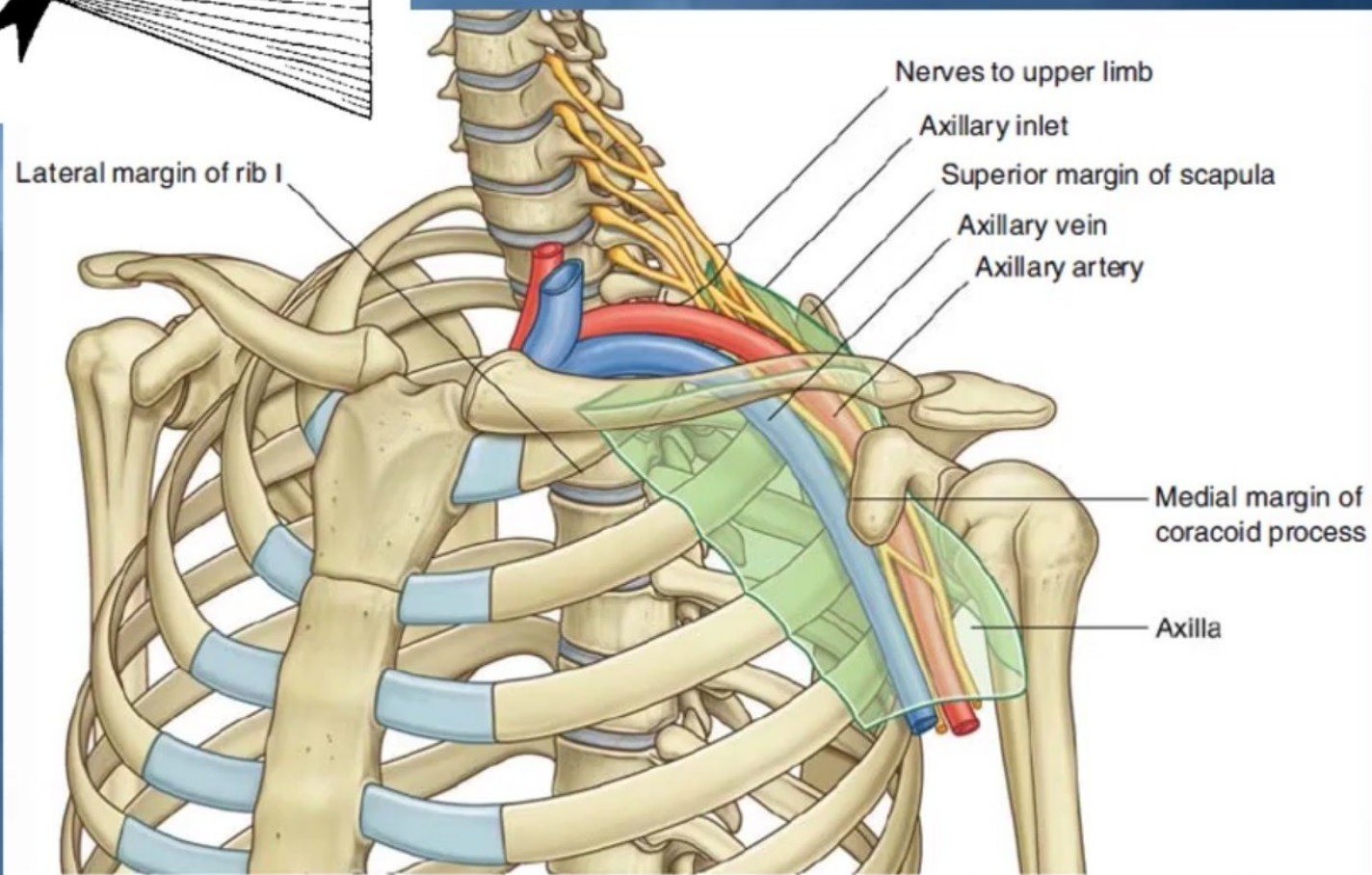
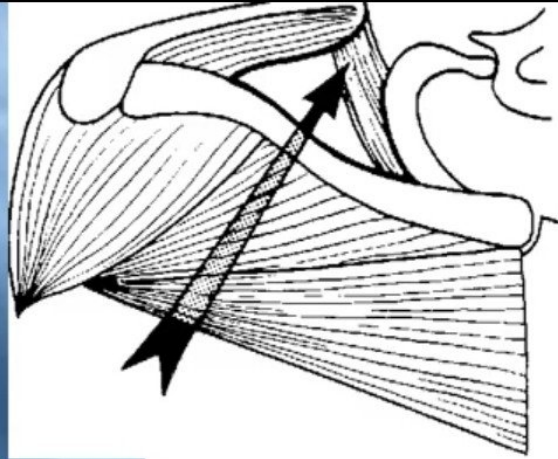
Dr. Osman Ahmed Osman

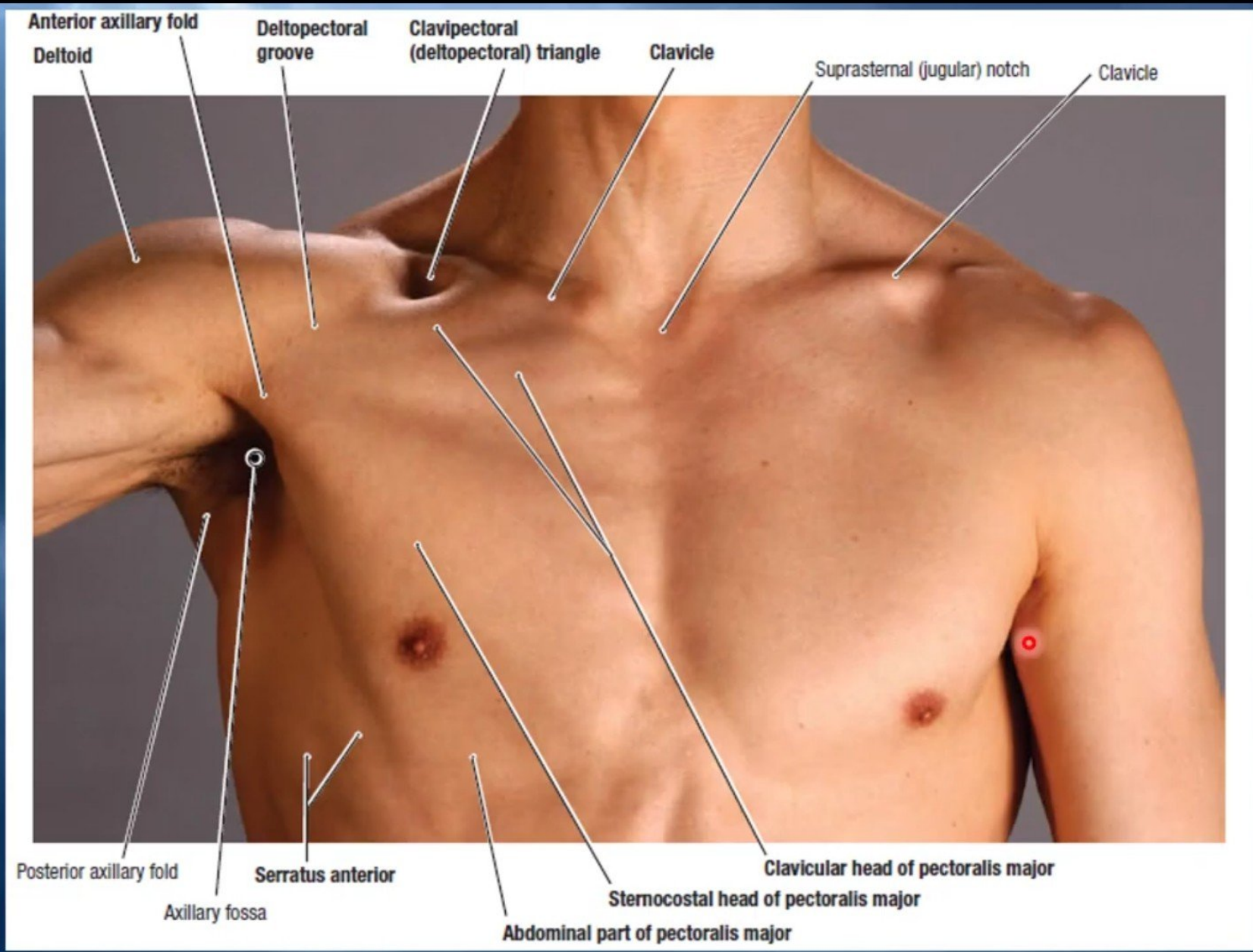
Lecture Points:

- Definition of axilla
- Axillary Boundaries
- Apex of the axilla.
- Base of the axilla.
- Axillary folds
- Anterior wall of the axilla.
- lateral wall of the axilla.
- Medial wall of the axilla.
- Posterior wall of the axilla.
- Gateways in the posterior wall.
- Over view of the muscles was mentioned in axillary walls.
- Contents of the Axilla.
- Axillary sheath.
- Axillary artery (relation).
- Branches of axillary artery
- Beginning and termination (End).
- Axillary Artery: divided into three parts.
- Palpation of the axillary artery
- Summary for axillary artery
- Axillary vein
- Axillary Lymph Nodes.
- Follow of lymphatic derange.

Definition of axilla

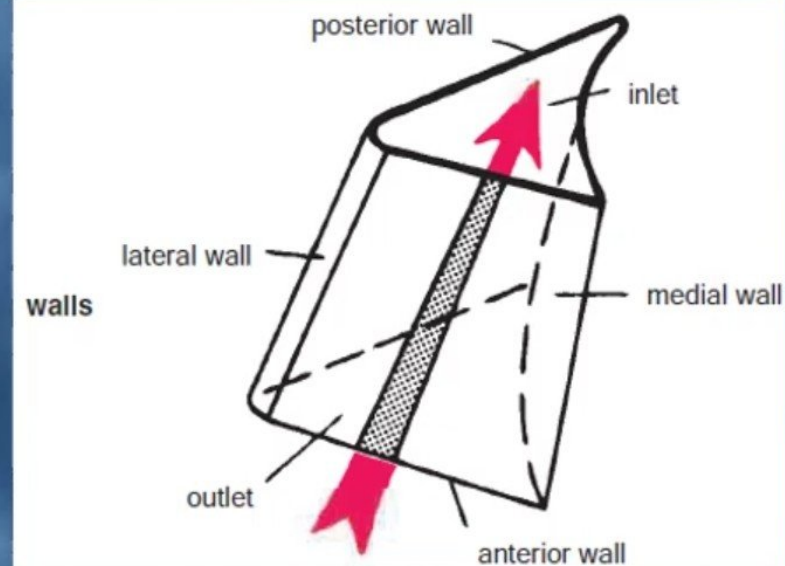
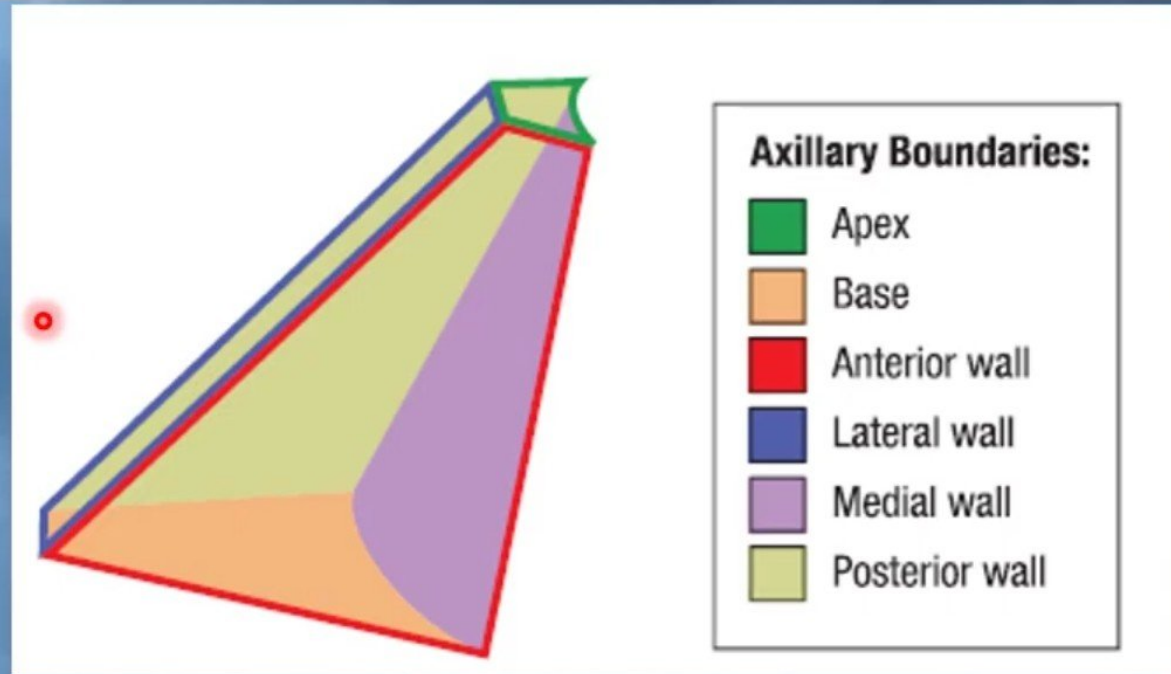
- Is the space between **the medial part of the arm** and **the side of the chest**.
- It is **a pyramid-shaped** space between the **upper thoracic wall** and **the arm**.
- It forms an important passage for **nerves** and **vessels** from the **neck** to the **upper limb**.

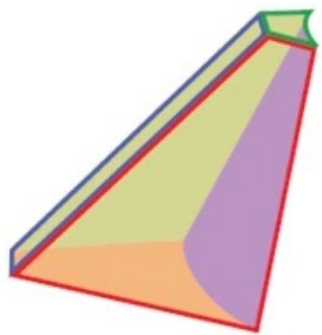




Axillary Boundaries

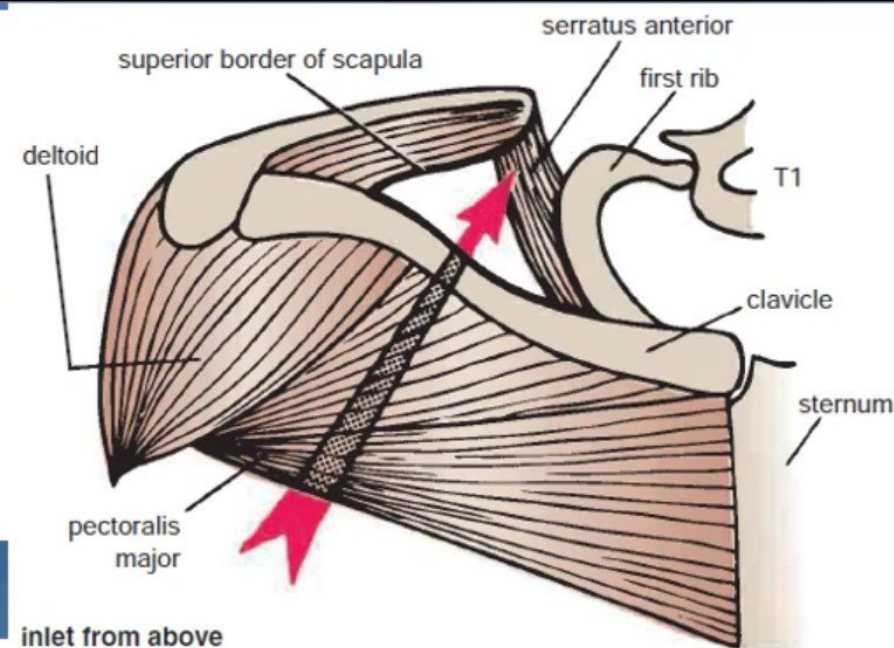
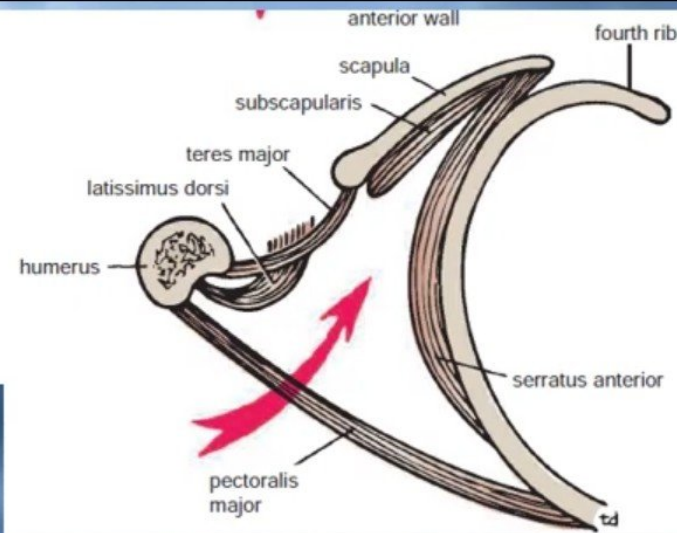
- Apex
- Base
- Anterior wall
- lateral wall
- Medial wall
- Posterior wall





Axillary Boundaries:

- Apex
- Base
- Anterior wall
- Lateral wall
- Medial wall
- Posterior wall



Boundaries of the Axilla

Apex: interval between the **clavicle**, **first rib**, and **upper border of the scapula**.

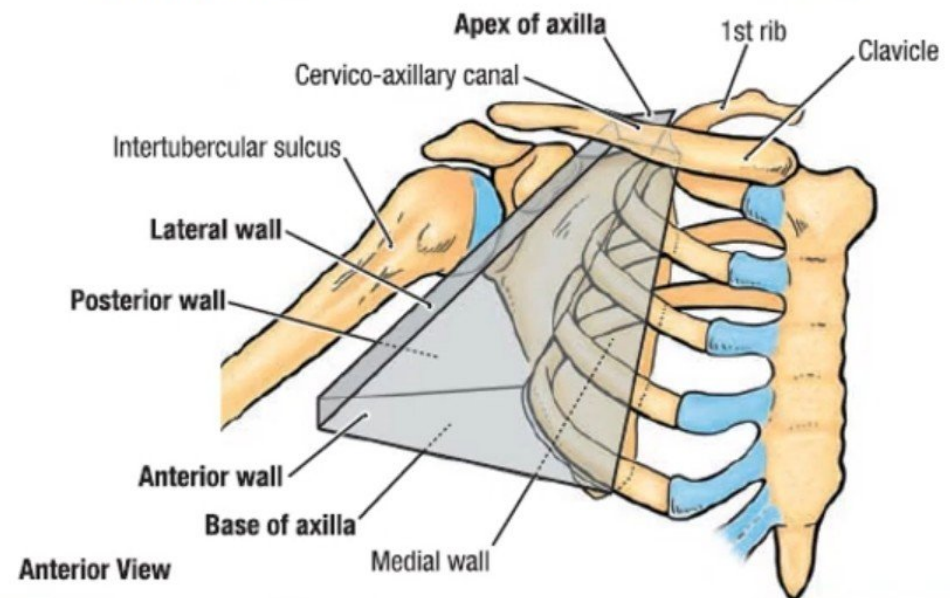
Base: axillary fascia and skin.

Anterior wall: **pectoralis major** and **pectoralis minor** muscles and **clavipectoral fascia**;

Medial wall: upper ribs and their **intercostal muscles** and **serratus anterior** muscle;

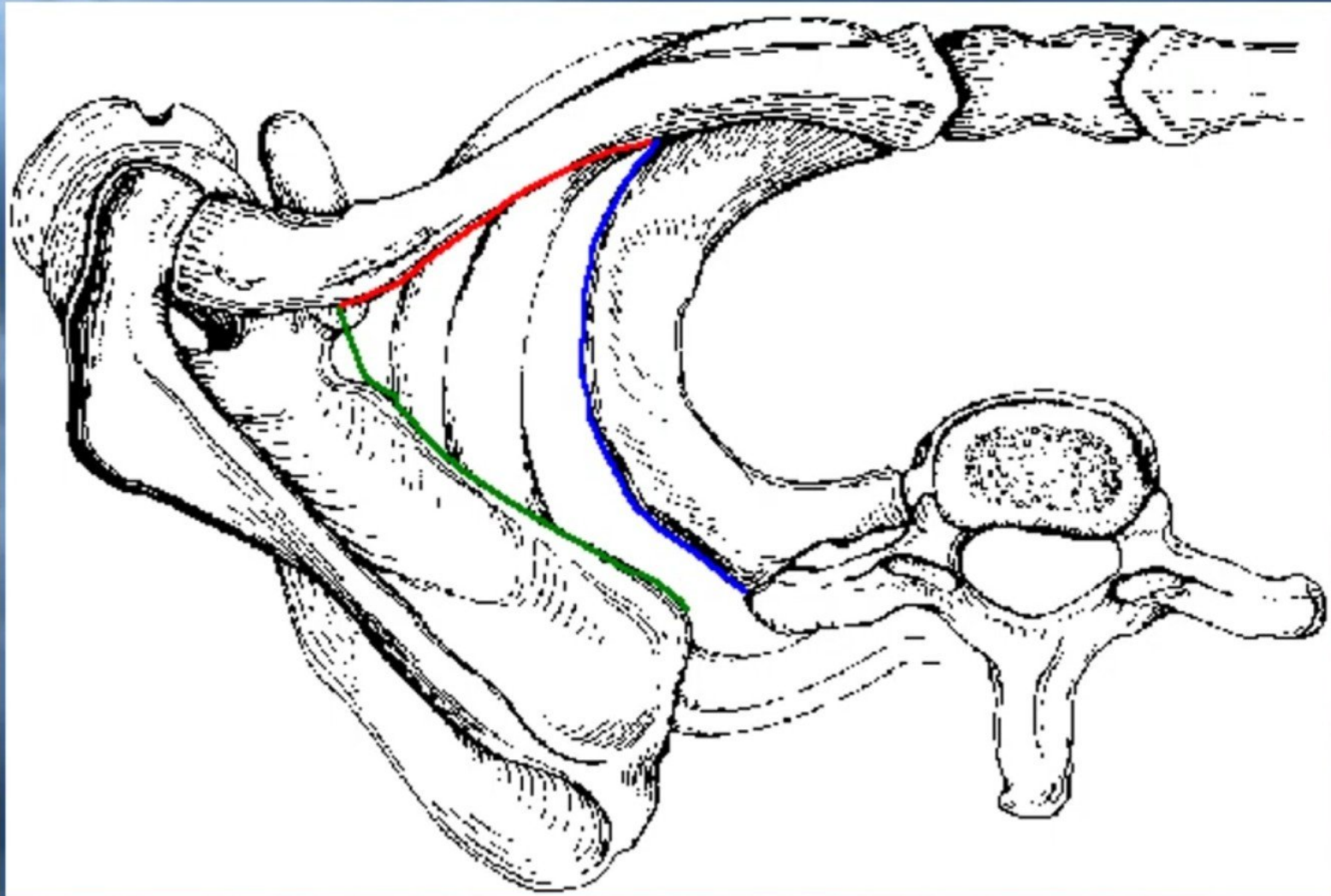
lateral wall: Intertubercular groove of the humerus;

Posterior wall: subscapularis, teres major, and latissimus dorsi muscles;



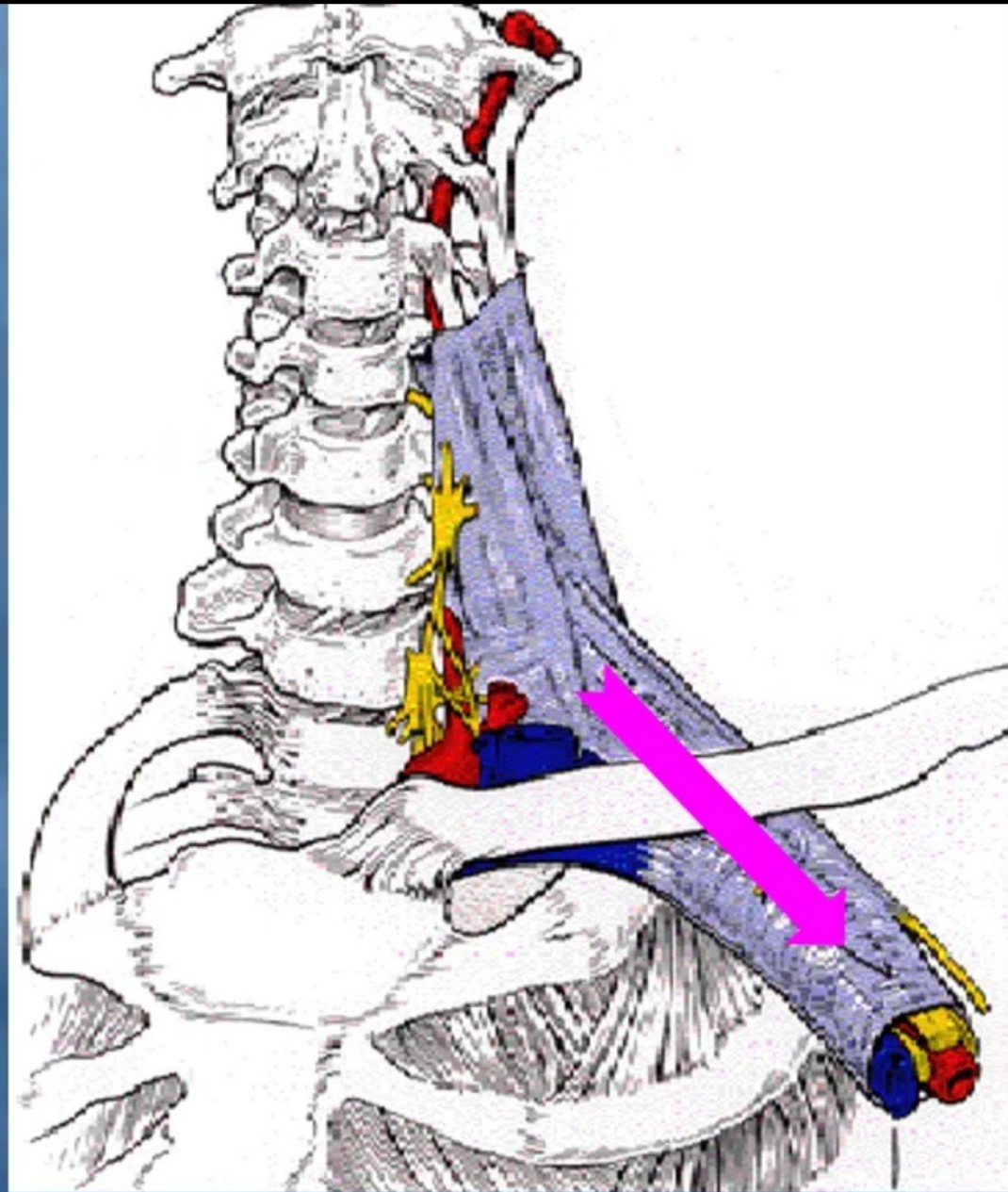
Apex of the axilla

- The apex is directed towards the root of the neck
- it is a narrow triangular gap which is bounded by the **clavicle anteriorly**, **scapula posteriorly**, and the **first rib medially**.



Apex of the axilla

- The apex is also called **the inlet** since it allows the entrance to the axilla of nerves and vessels.

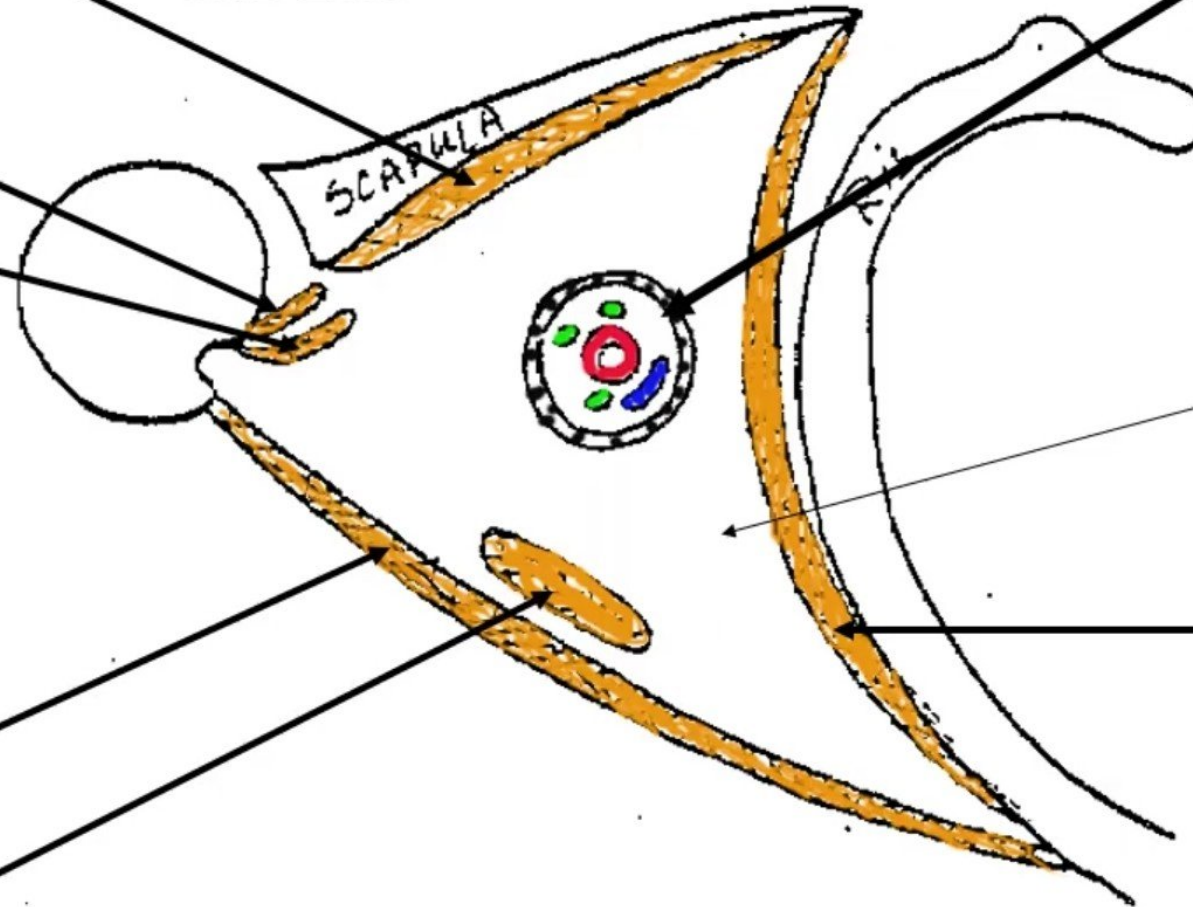


Muscles in inlet of the axilla

Major structures inside: Axillary sheath and contents!

Most of the rest of the space is adipose tissue

Serratus anterior



Subscapularis

Latissimus dorsi

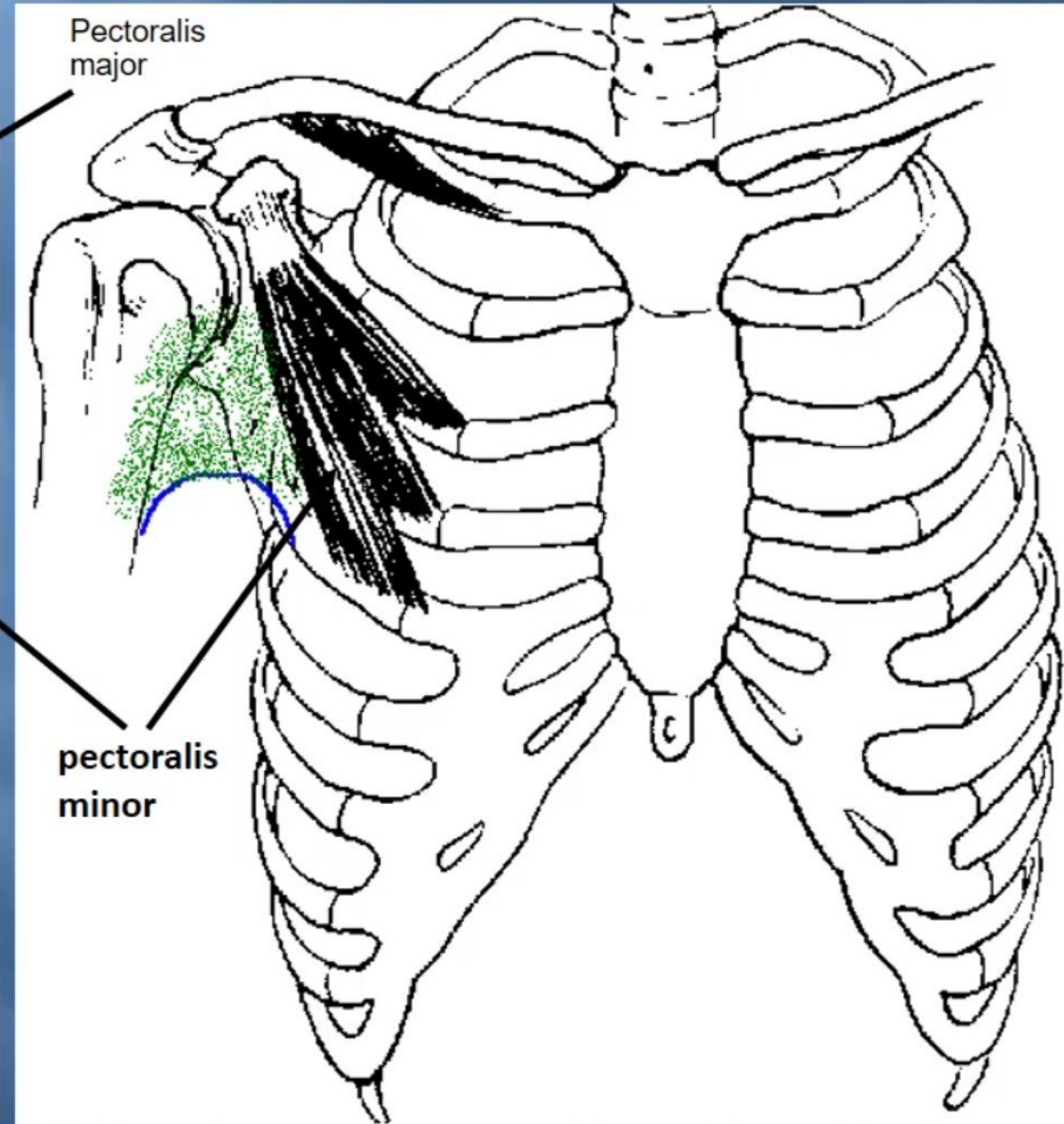
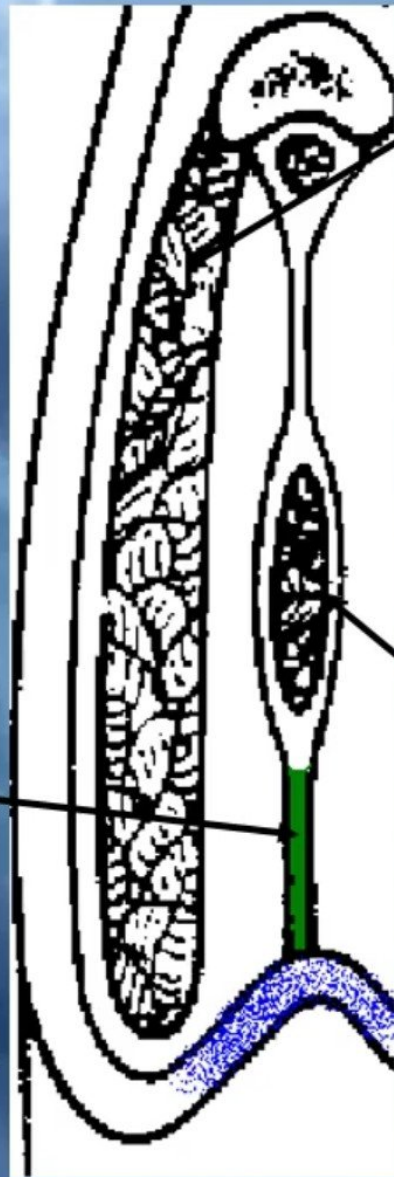
Teres major

Pectoralis major

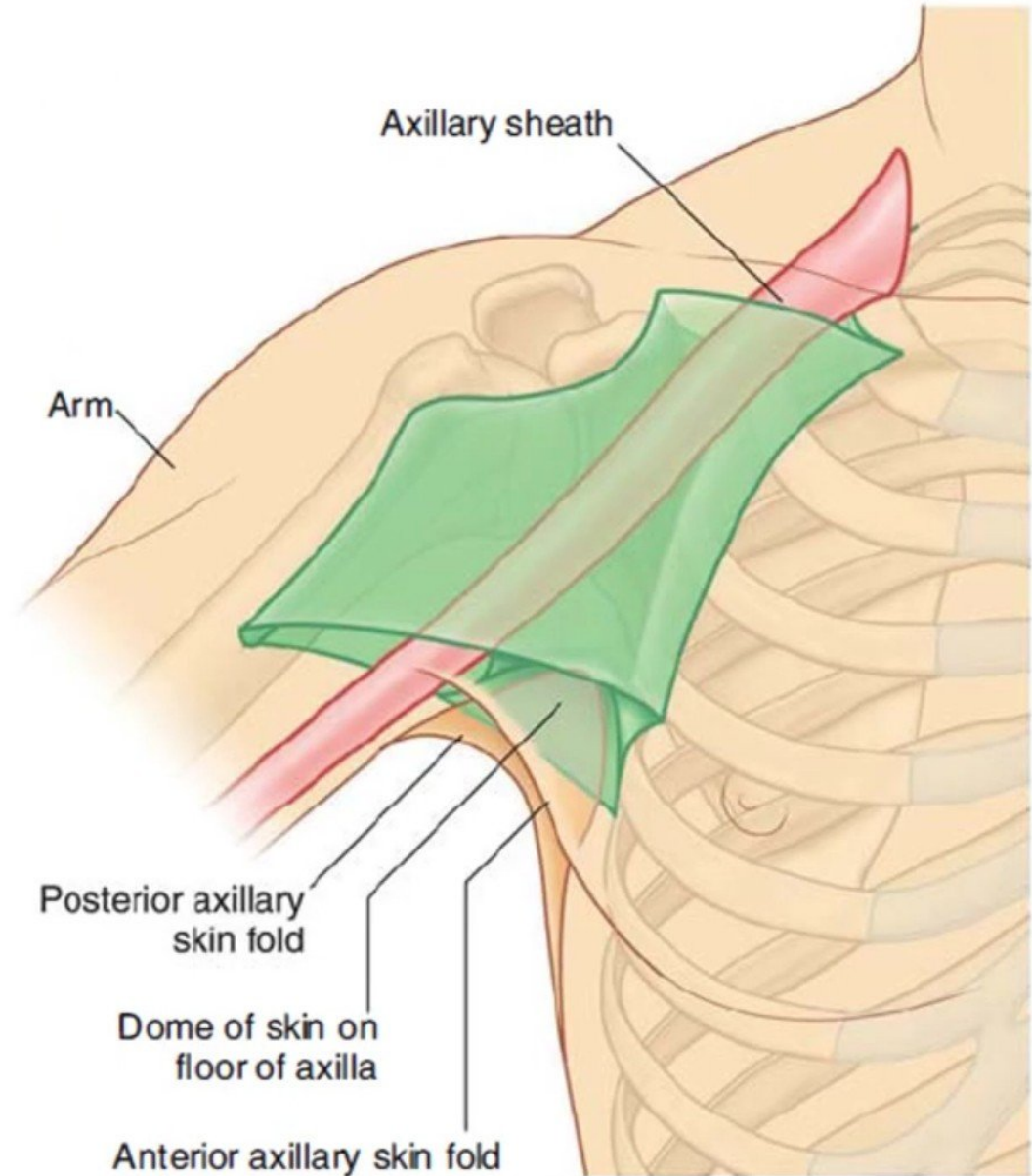
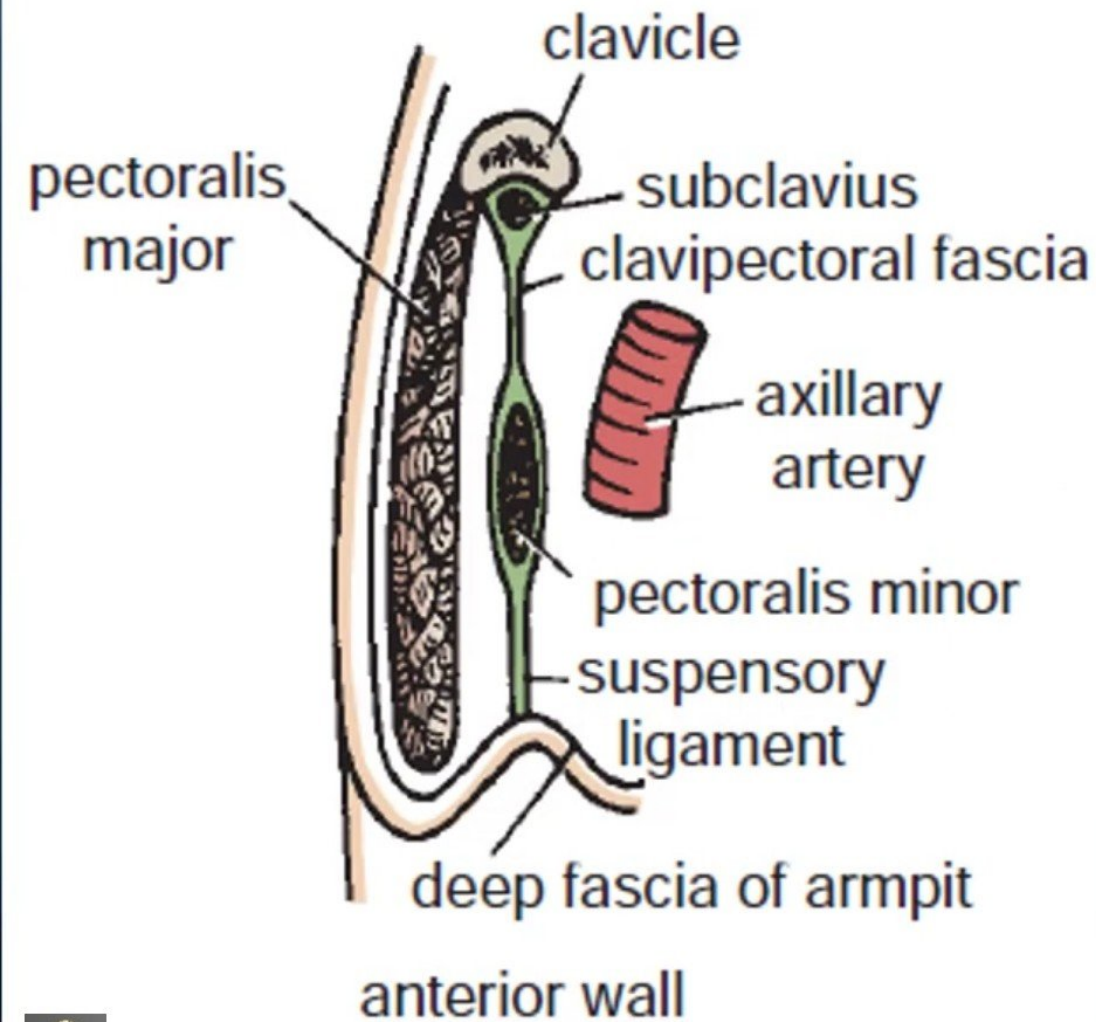
Pectoralis minor

Base of the axilla

- The base faces inferiorly and is formed by **fascia** and **skin**.
- The fascia is supported by the **suspensory ligament of the axilla** which renders the base hollow

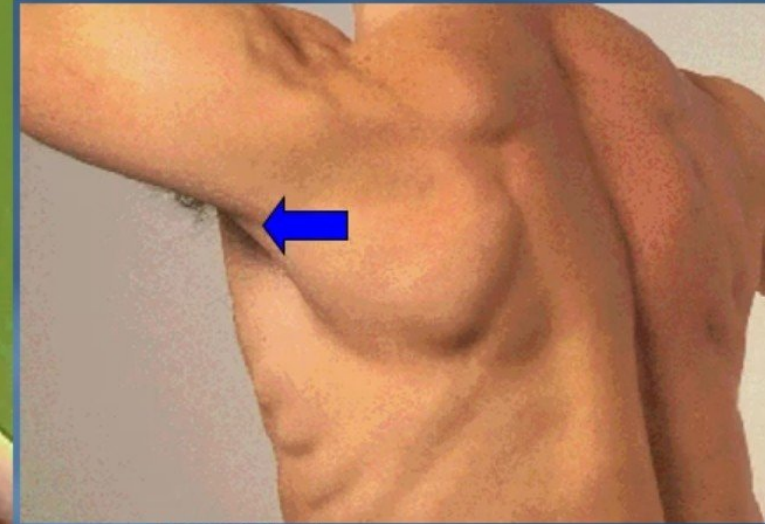
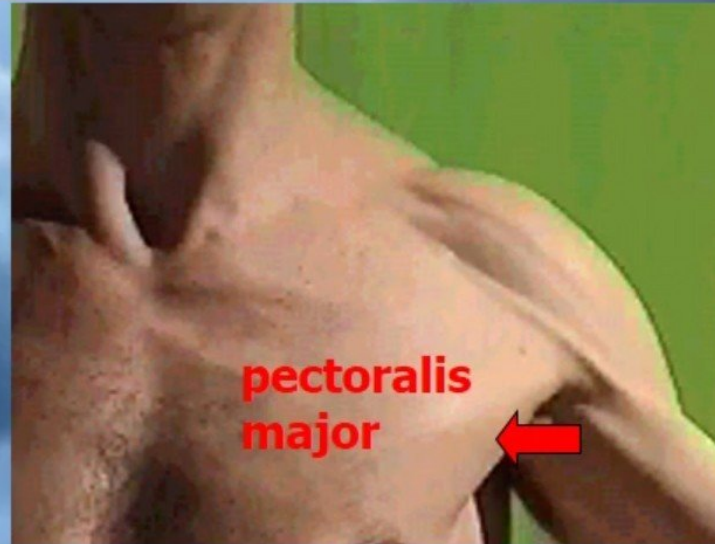


Base (Floor) of the axilla



Axillary folds

- The anterior and posterior walls of the axilla can be palpated by fingers and thumb thus forming the anterior and posterior axillary folds.



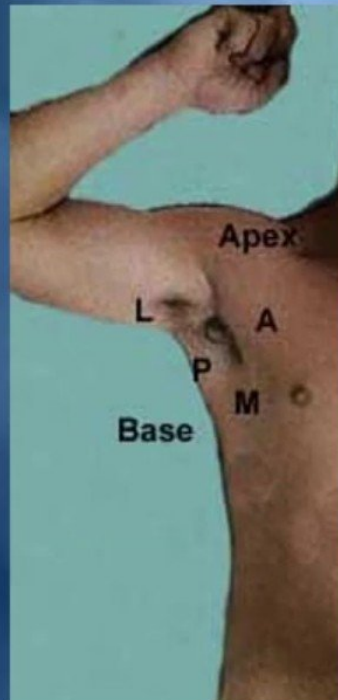
Axillary folds:

Anterior axillary fold:

- Pectoralis major

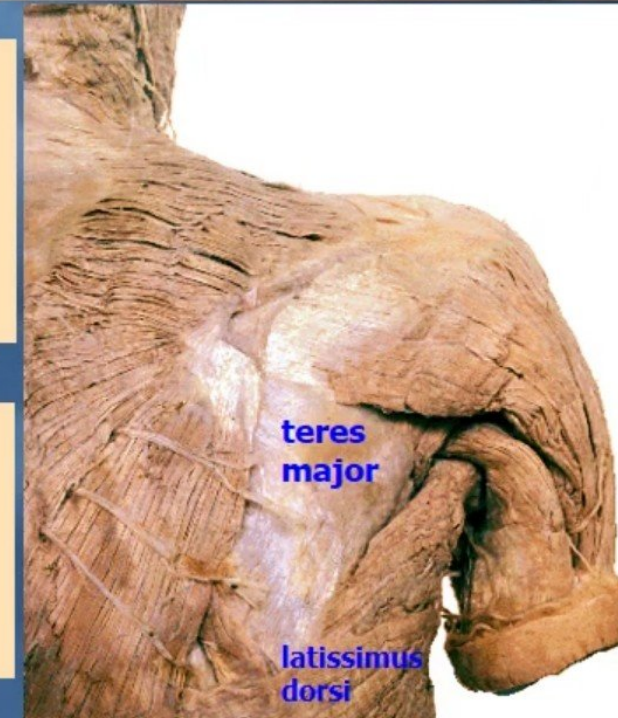
Posterior axillary fold:

- Latissimus dorsi
- Teres major



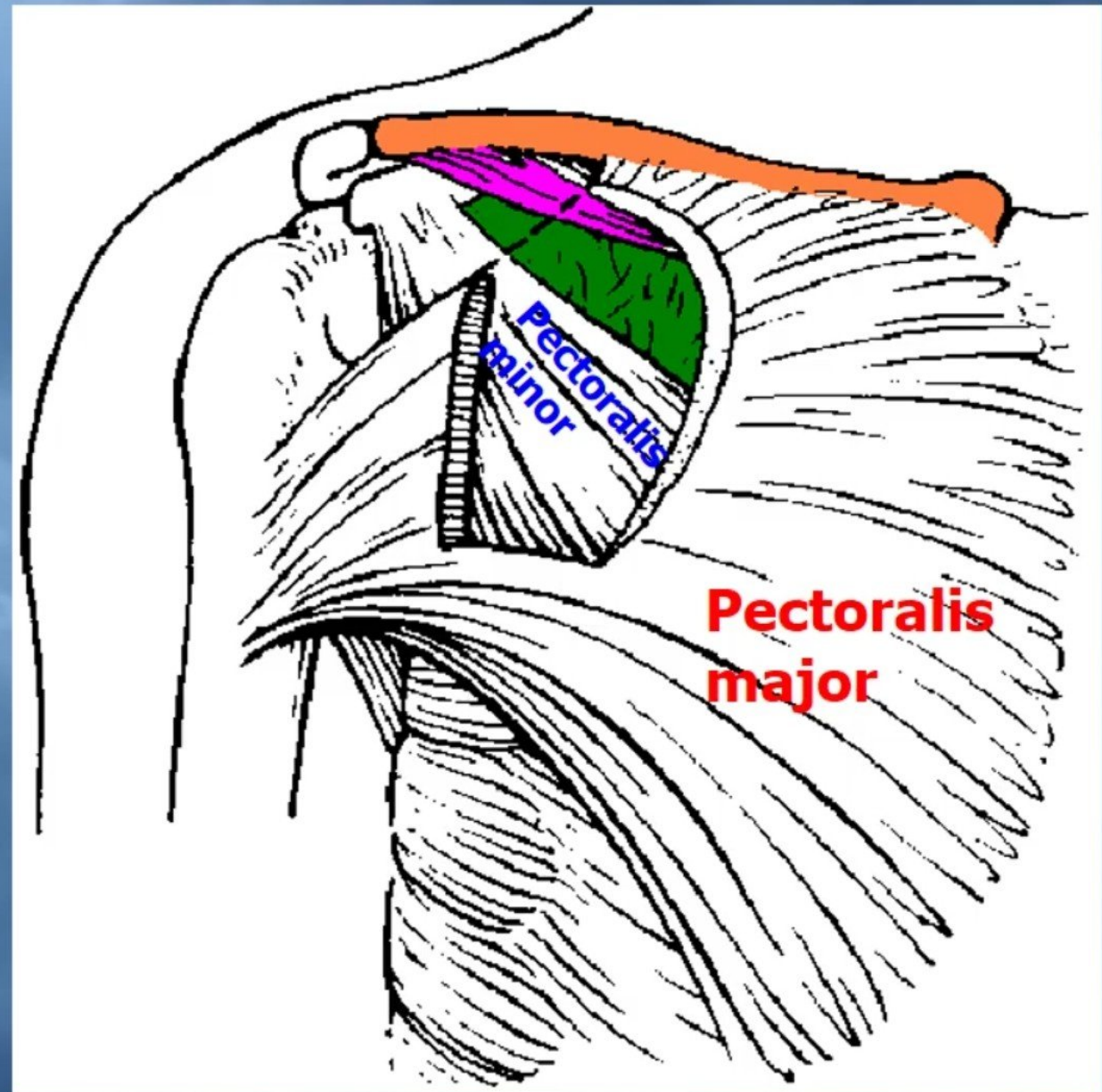
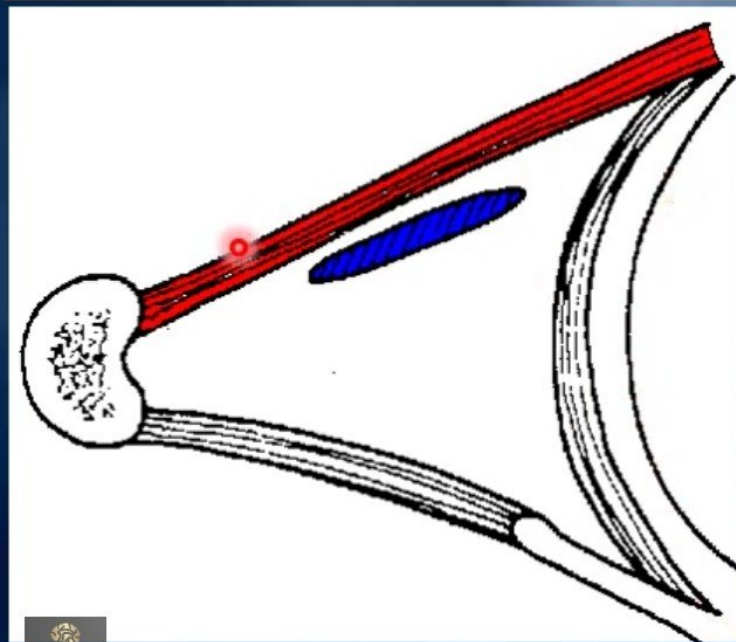
⌘ The **anterior axillary fold** is formed by the lateral border of pectoralis major.

⌘ The **posterior axillary fold** is formed by teres major and latissimus dorsi



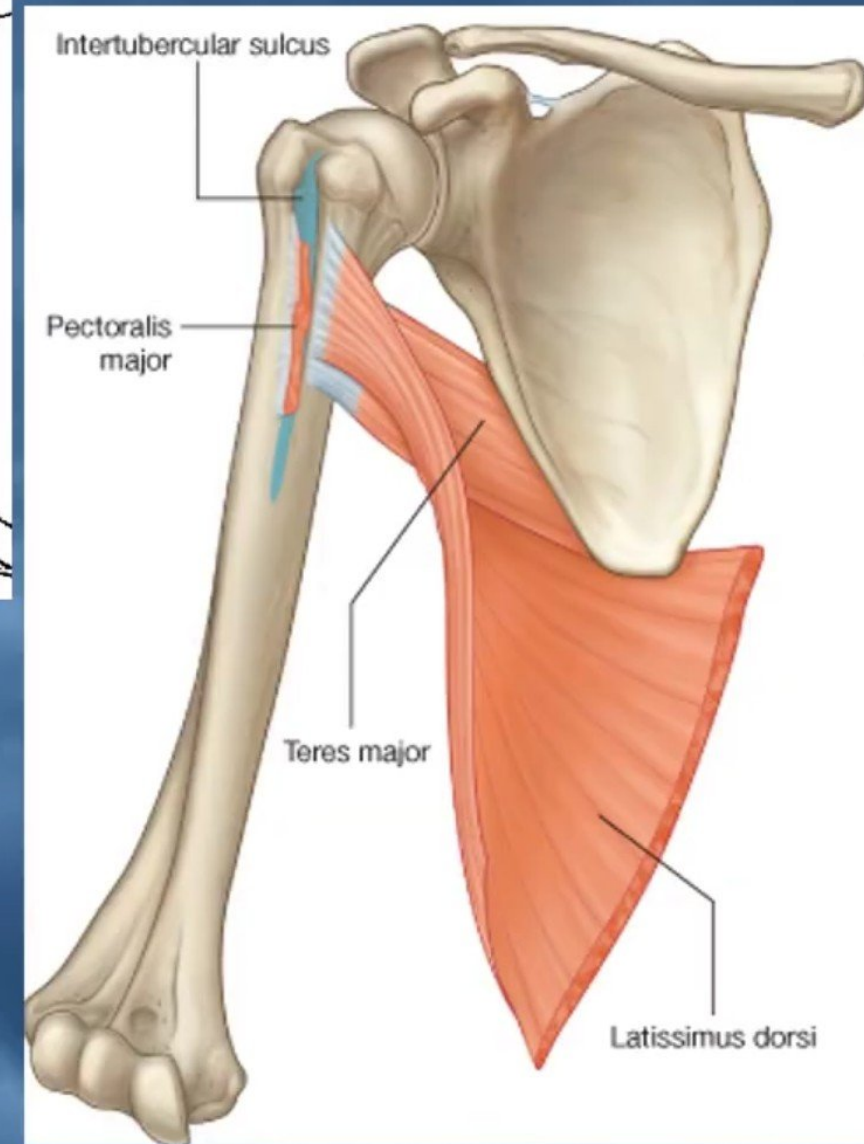
Anterior wall of the axilla

- This is formed by the clavicle, pectoralis major, pectoralis minor, subclavius, and the clavipectoral fascia

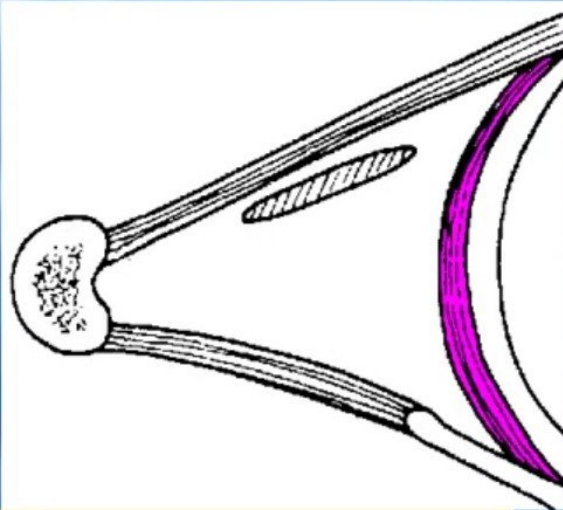


Lateral wall of axilla

- The lateral wall is narrow and is formed by the floor of the intertubercular groove of the humerus to which is attached the tendon of **latissimus dorsi** and in which runs the tendon of the **long head of biceps**
- Bicipital groove/intertubercular sulcus of the humerus.
- Coraco brachialis
- Biceps brachii.

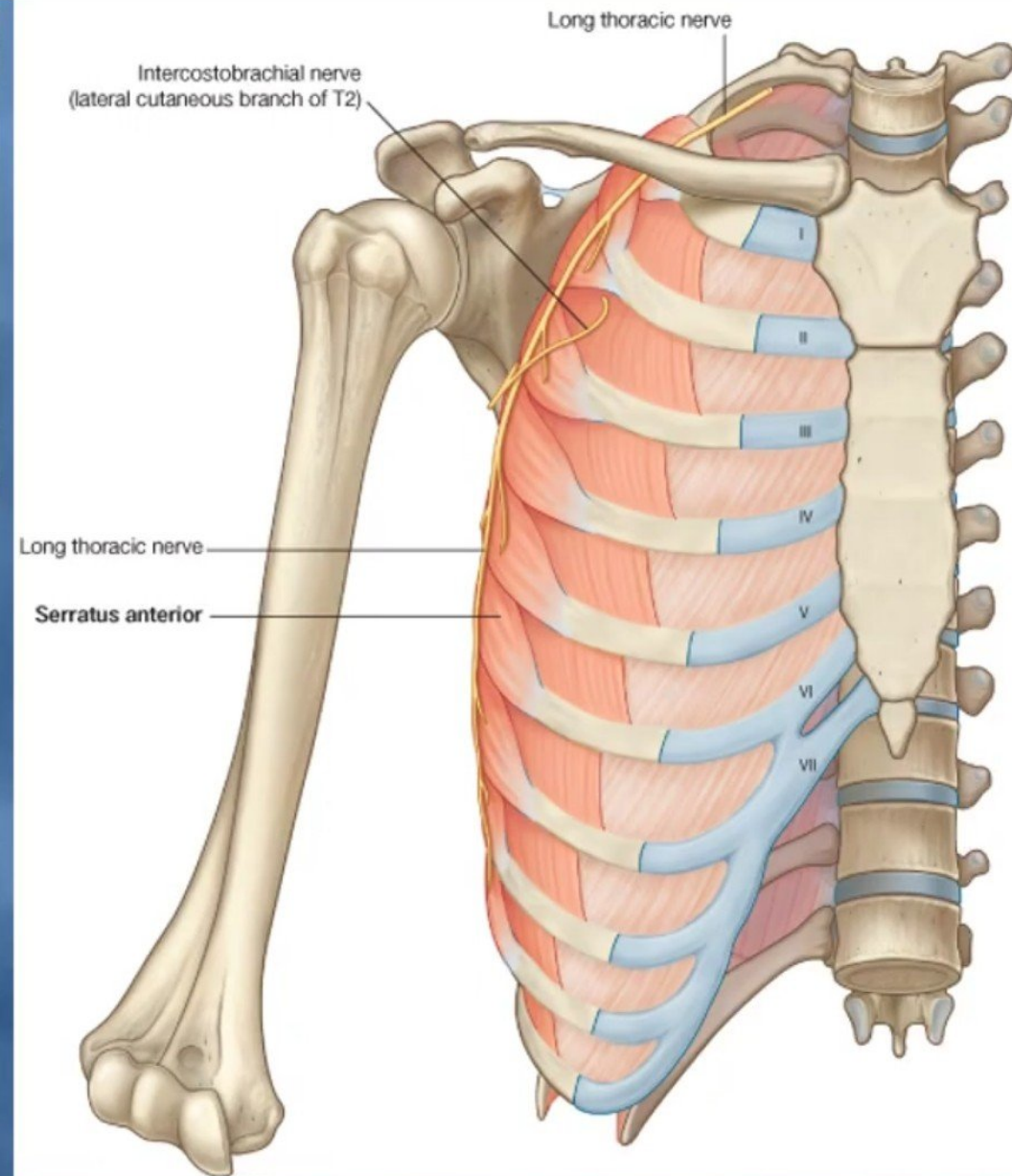


Medial wall of axilla



The medial wall is formed by:

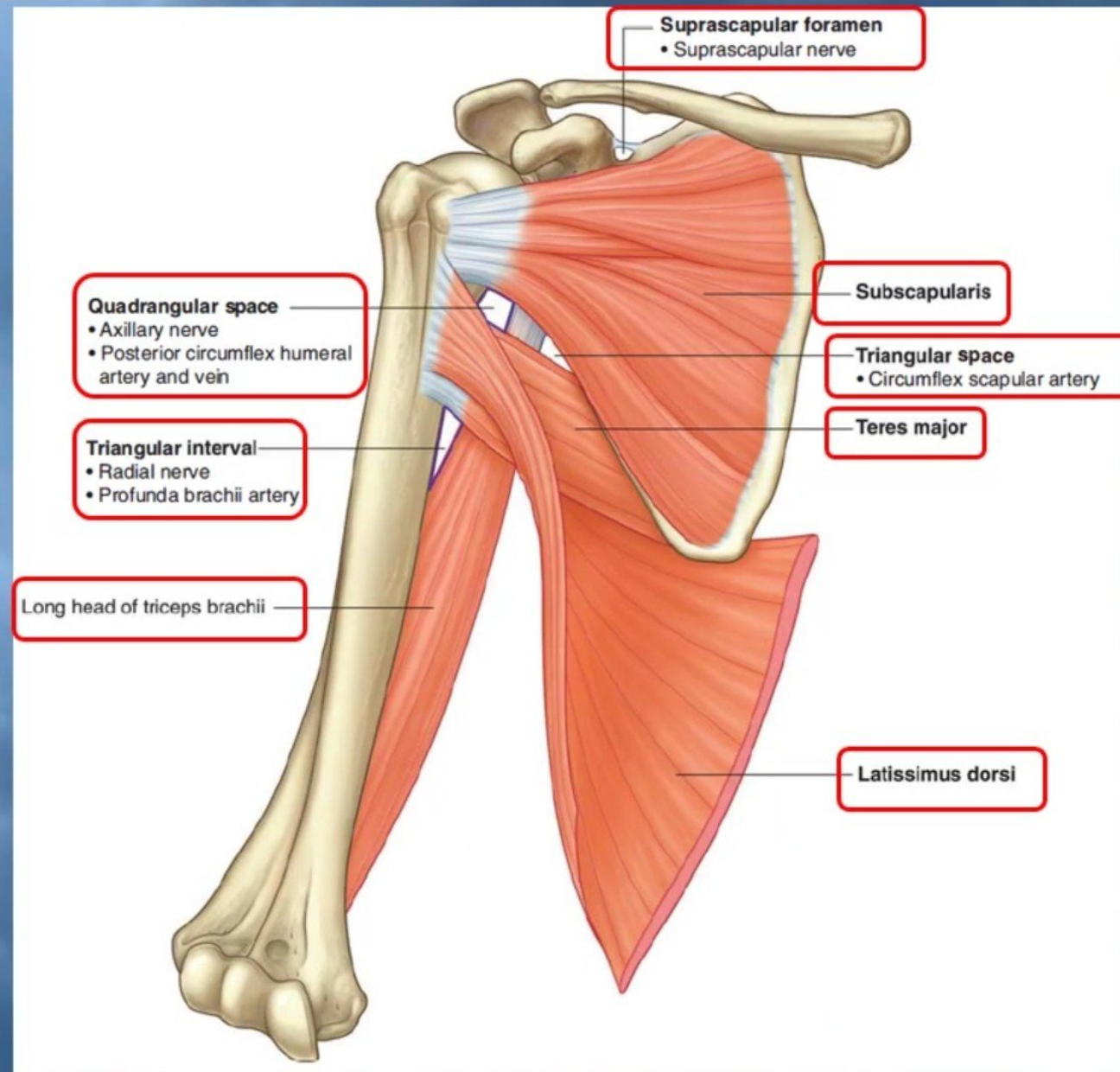
- The upper ribs.
- upper 3 or 4 intercostal spaces.
- upper part of serratus anterior.
- long thoracic nerve.
- intercosto brachial nerve pierces the medial wall.



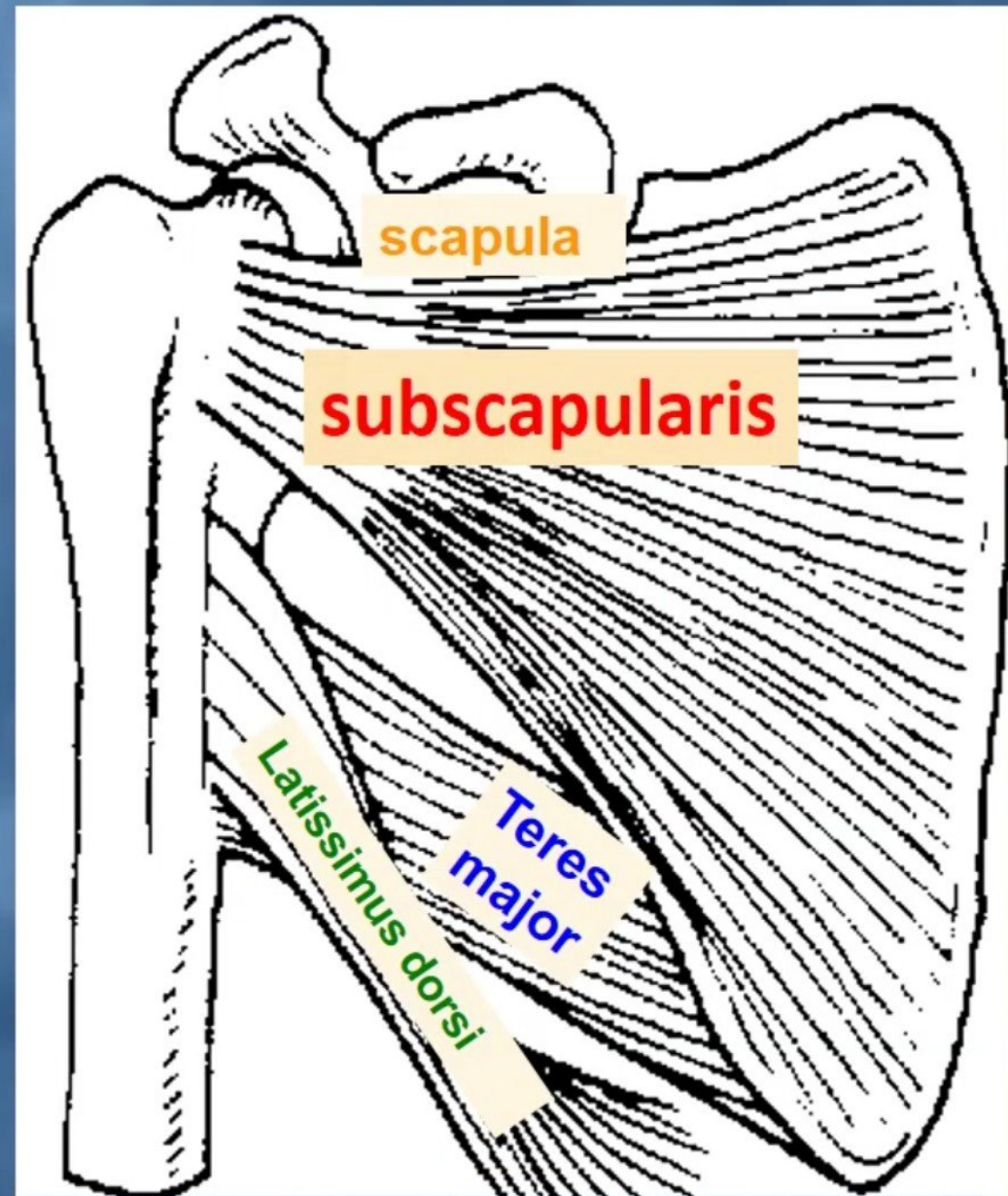
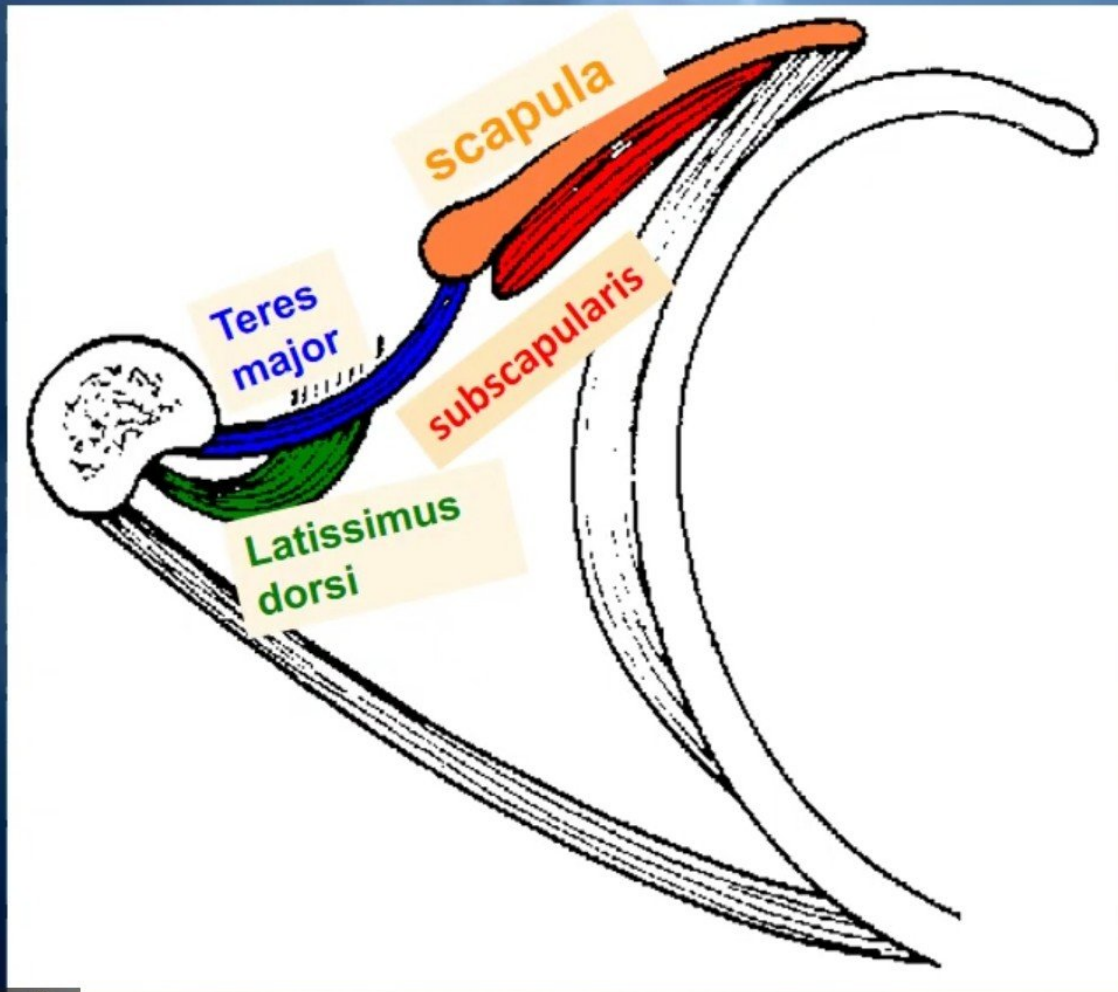
Posterior wall of axilla

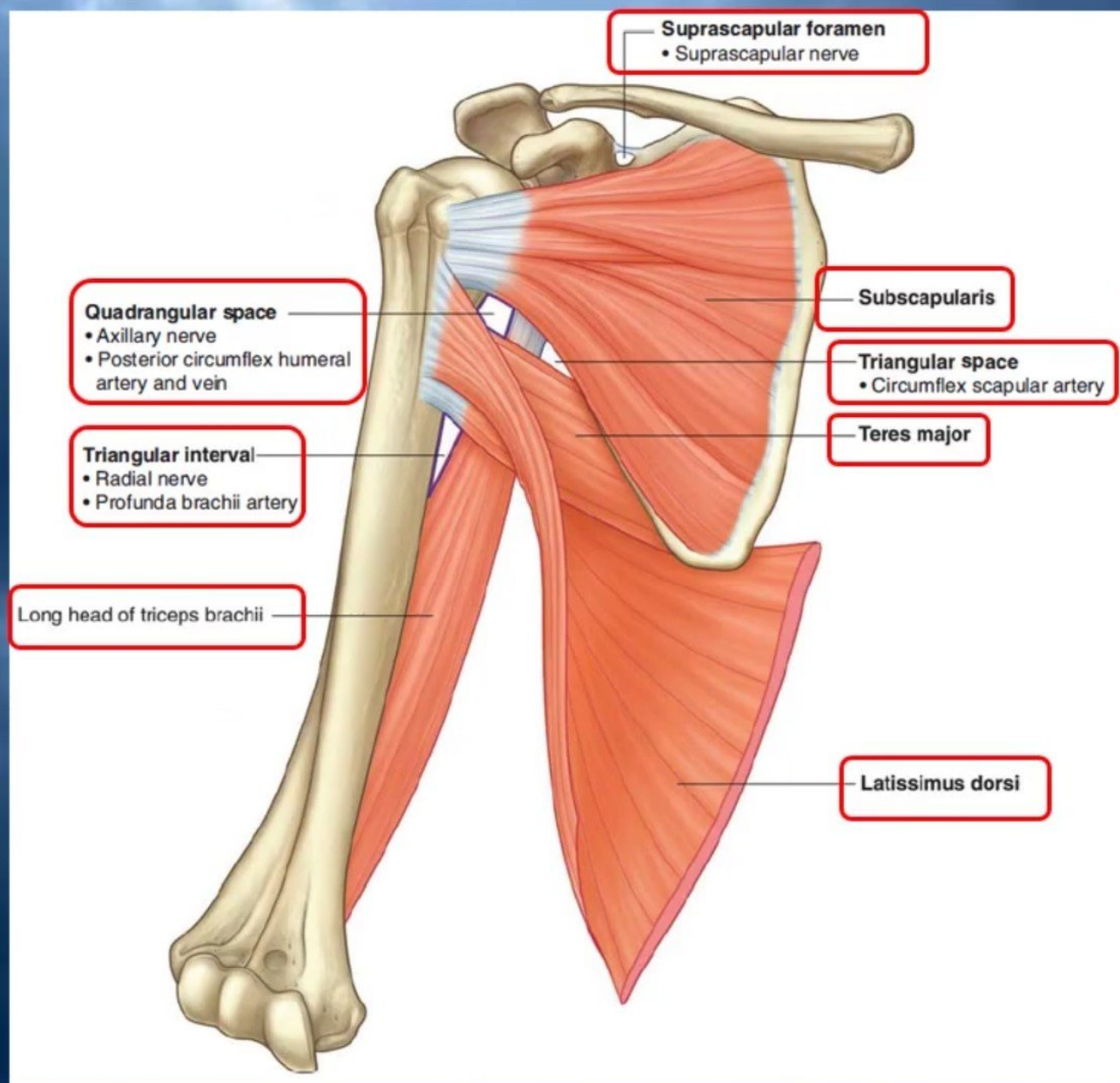
Structures in the posterior wall:

- scapula
- Subscapularis
- Teres major
- Latissimus dorsi (forms posterior axillary fold)



Posterior wall of the axilla



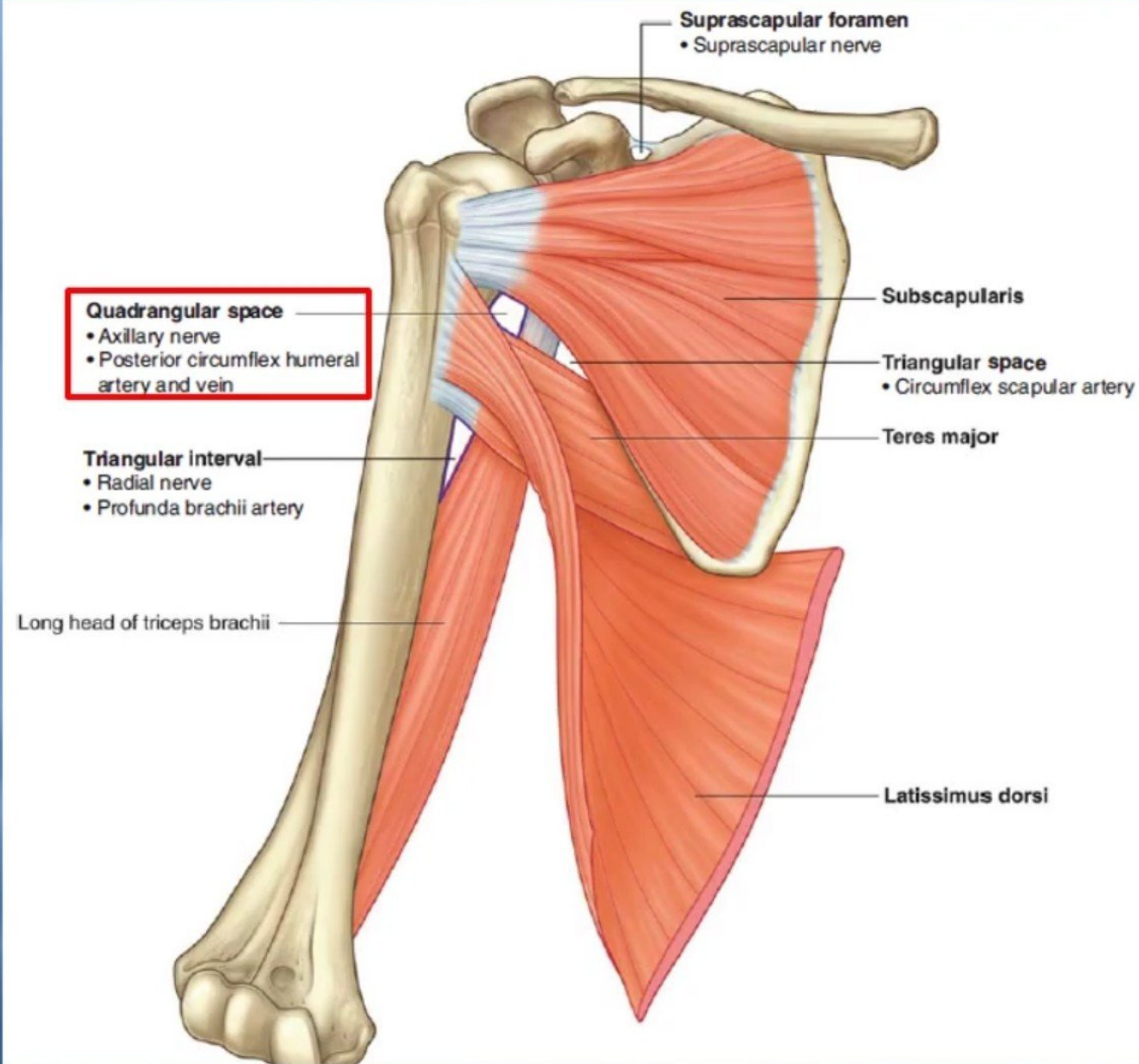


Posterior wall of axilla

Gateways in the posterior wall

Quadrangular space

- provides a passageway for nerves and vessels passing between the axilla and the more posterior scapular and deltoid regions
- **Boundaries are formed by:**
 - The inferior margin of the **subscapularis/teres minor** muscles
 - The **surgical neck of the humerus**;
 - The **superior margin of the teres major muscle**.
- **Axillary nerve and posterior circumflex humeral artery and vein** passes through it

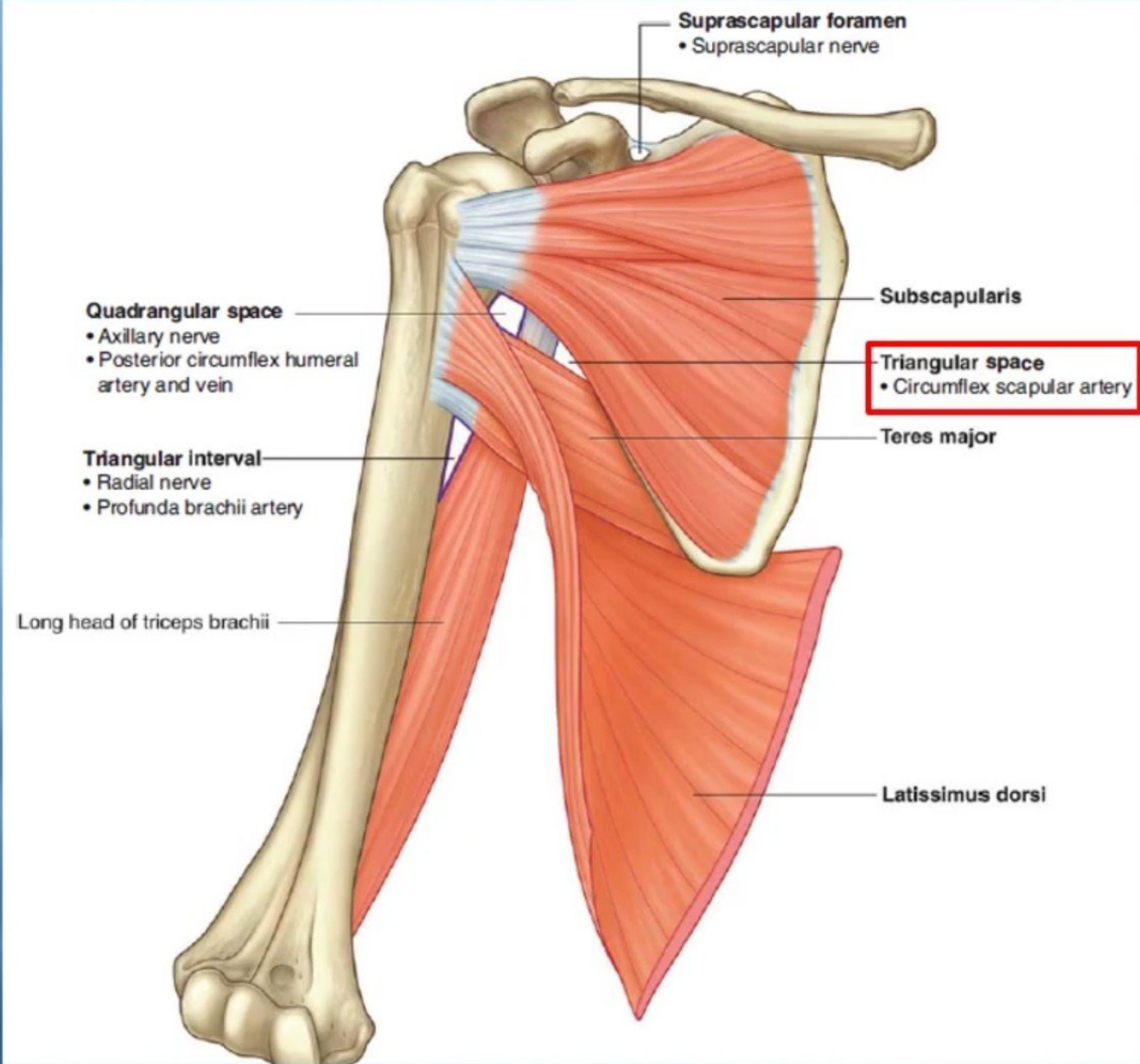


Gateways in the posterior wall

Triangular spaces

Upper triangular space:

- When viewed from anteriorly,
- it is formed
- by:
 - Medial margin of the **long head of the triceps brachii**
 - Superior margin of the teres major muscle.
 - Inferior margin of the subscapularis/teres minor muscles.
- **Circumflex scapular artery & vein** pass into this space.

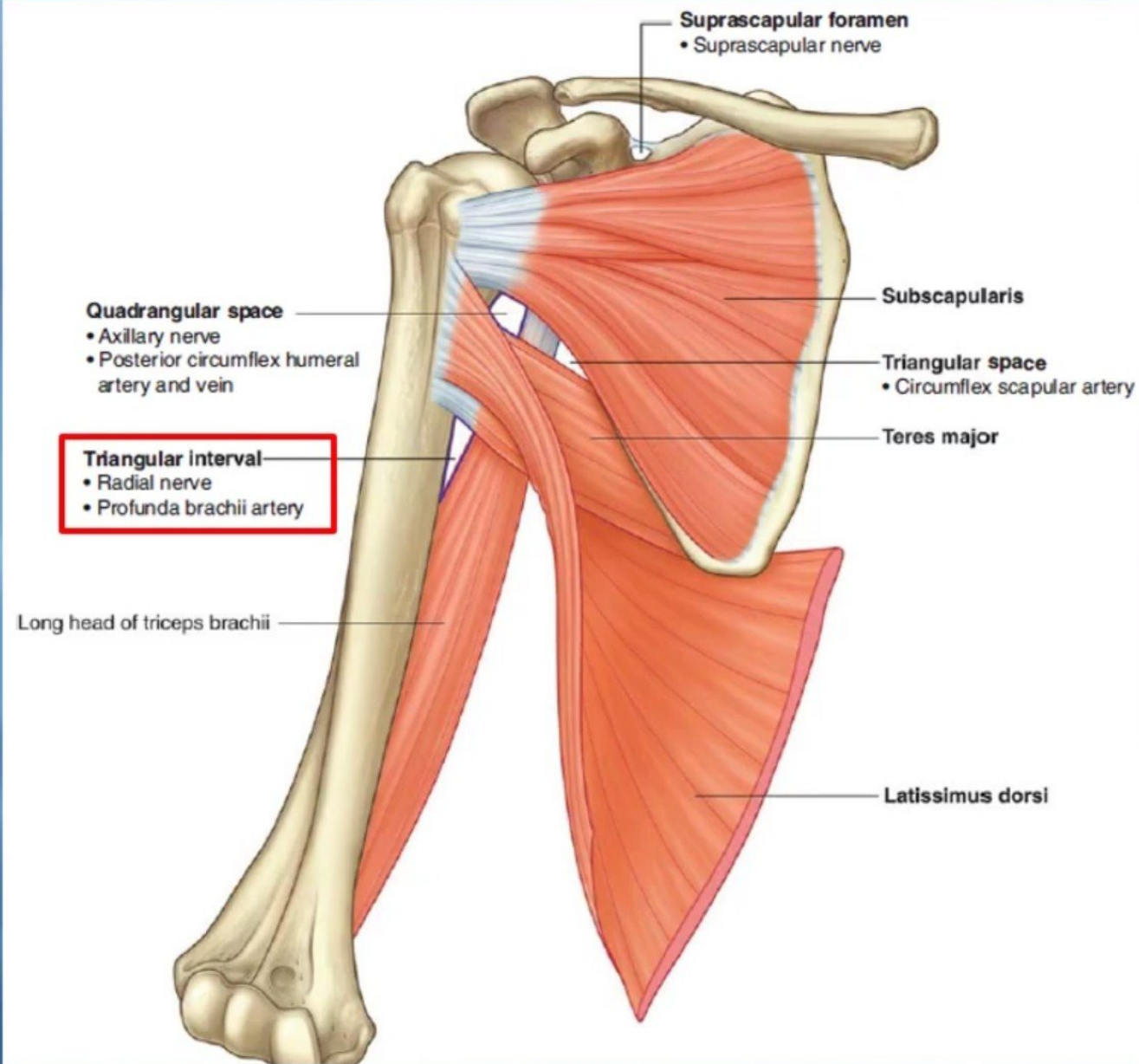


Gateways in the posterior wall

Lower triangular interval

It is formed by:

- Long head of the triceps brachii muscle.
- The shaft of the humerus.
- The inferior margin of the teres major muscle.
- The **radial nerve** passes out of the axilla traveling through this interval to reach the posterior compartment of the arm.



Over view of the muscles was mentioned in axillary walls

Table Muscle of the medial wall of the axilla (spinal segment in bold is the major segment innervating the muscle)

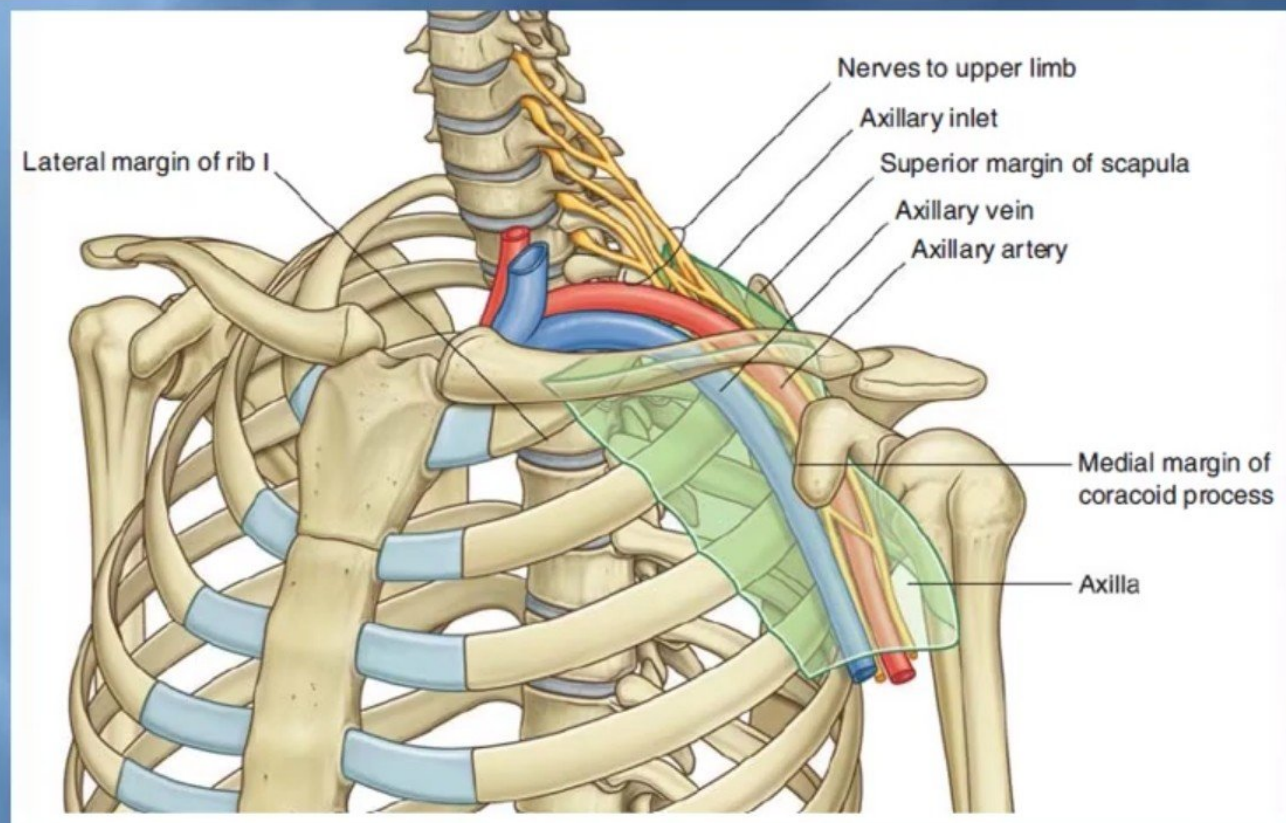
Muscle	Origin	Insertion	Innervation	Function
Serratus anterior	Lateral surfaces of upper 8–9 ribs and deep fascia overlying the related intercostal spaces	Costal surface of medial border of scapula	Long thoracic nerve (C5, C6, C7)	Protraction and rotation of the scapula; keeps medial border and inferior angle of scapula opposed to thoracic wall

Table Muscles of the lateral and posterior wall of the axilla (spinal segments in bold are the major segments innervating the muscle; spinal segments in parentheses do not consistently innervate the muscle)

Muscle	Origin	Insertion	Innervation	Function
Subscapularis	Medial two-thirds of subscapular fossa	Lesser tubercle of humerus	Upper and lower subscapular nerves (C5, C6, (C7))	Rotator cuff muscle ; medial rotation of the arm at the glenohumeral joint
Teres major	Elongate oval area on the posterior surface of the inferior angle of the scapula	Medial lip of the intertubercular sulcus on the anterior surface of the humerus	Lower subscapular nerve (C5, C6, C7)	Medial rotation and extension of the arm at the glenohumeral joint
Latissimus dorsi	Spinous processes of lower six thoracic vertebrae and related interspinous ligaments; via the thoracolumbar fascia to the spinous processes of the lumbar vertebrae, related interspinous ligaments, and iliac crest; lower 3–4 ribs	Floor of intertubercular sulcus	Thoracodorsal nerve (C6, C7, C8)	Adduction, medial rotation, and extension of the arm at the glenohumeral joint
Long head of triceps brachii	Infraglenoid tubercle on scapula	Common tendon of insertion with medial and lateral heads on the olecranon process of ulna	Radial nerve (C6, C7, C8)	Extension of the forearm at the elbow joint; accessory adductor and extensor of the arm at the glenohumeral joint

Contents of the Axilla

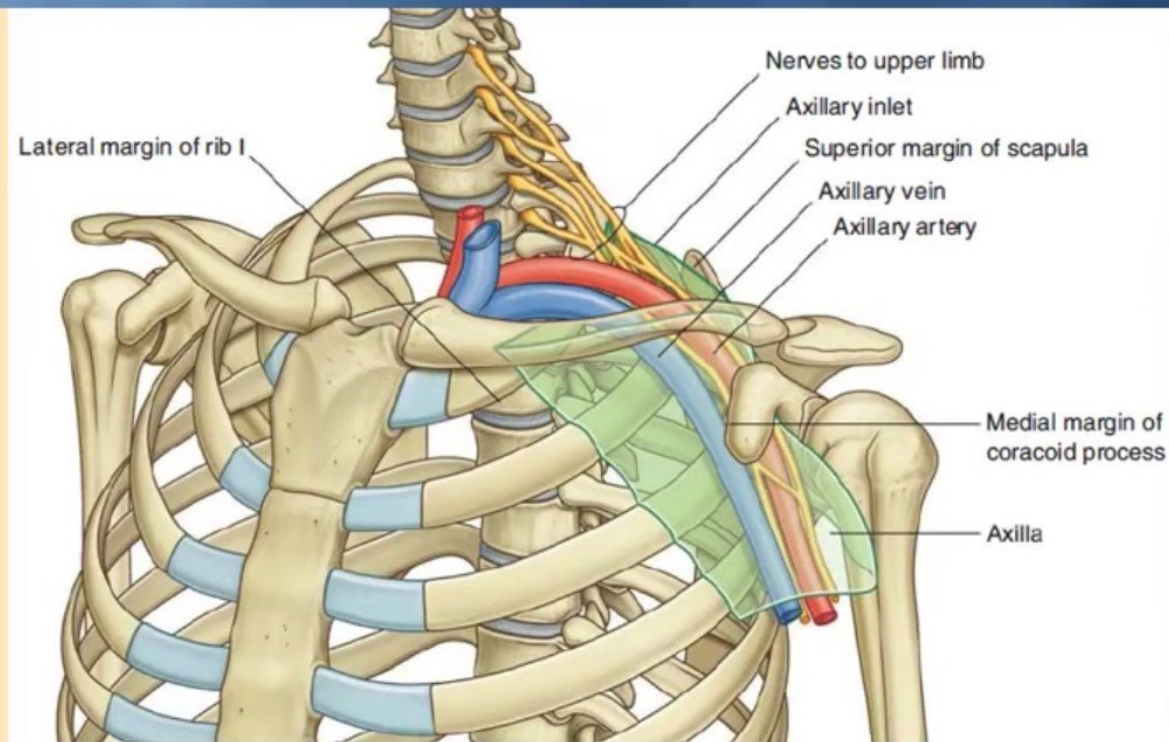
1. **Brachial plexus** and its branches.
2. **Axillary artery** has many branches, including the superior thoracic, thoracoacromial, lateral thoracic, thoracodorsal, and circumflex humeral (anterior and posterior) arteries.
3. **Axillary vein** is formed by the union of the brachial veins (venae comitantes of the brachial artery) and the basilic vein, receives the cephalic vein and veins that correspond to the branches of the axillary artery, and drains into the subclavian vein.



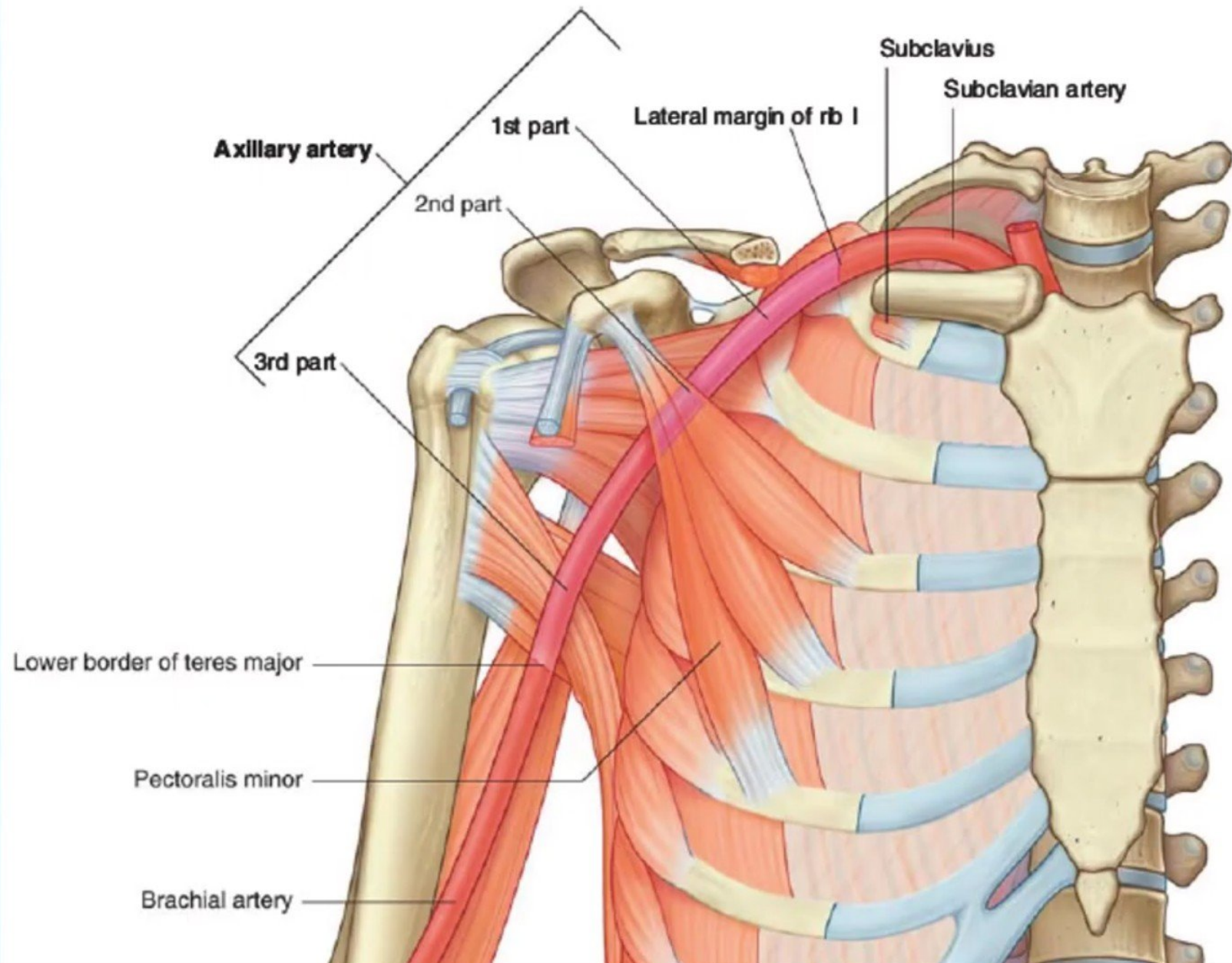
4. **Lymph nodes** and areolar tissue are present.
5. **Axillary tail** (tail of Spence) is a superolateral extension of the mammary gland.

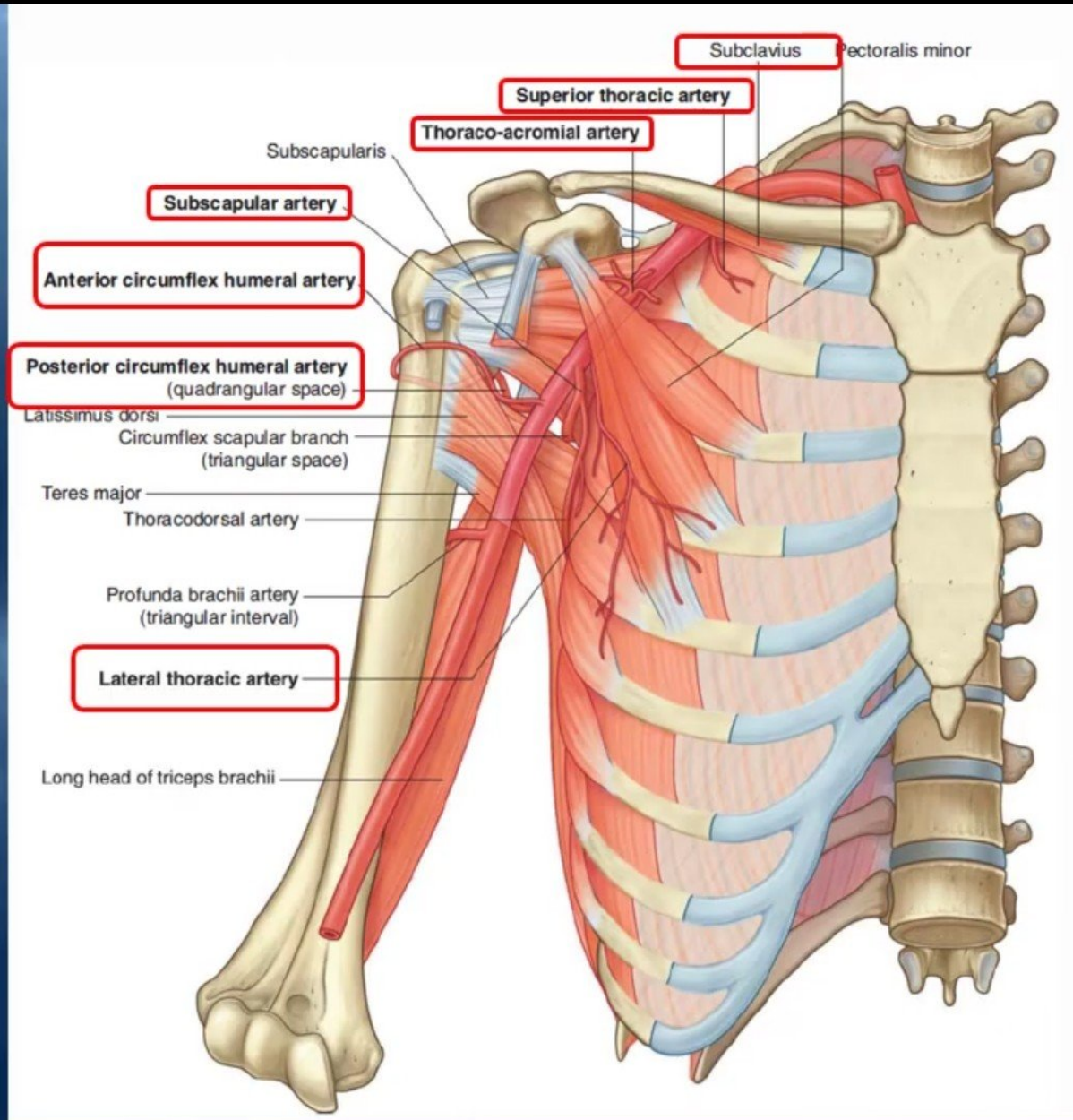
Axillary sheath

- Derived, at least in part, from anterior and middle scalene muscle fascia.
- Covers over a series of contents:
 - 1) **Axillary artery**
 - 2) **Axillary vein**
 - 3) **Brachial plexus** and nerves derived from it.
- The **axillary sheath** is just the fascia surrounding these structures.

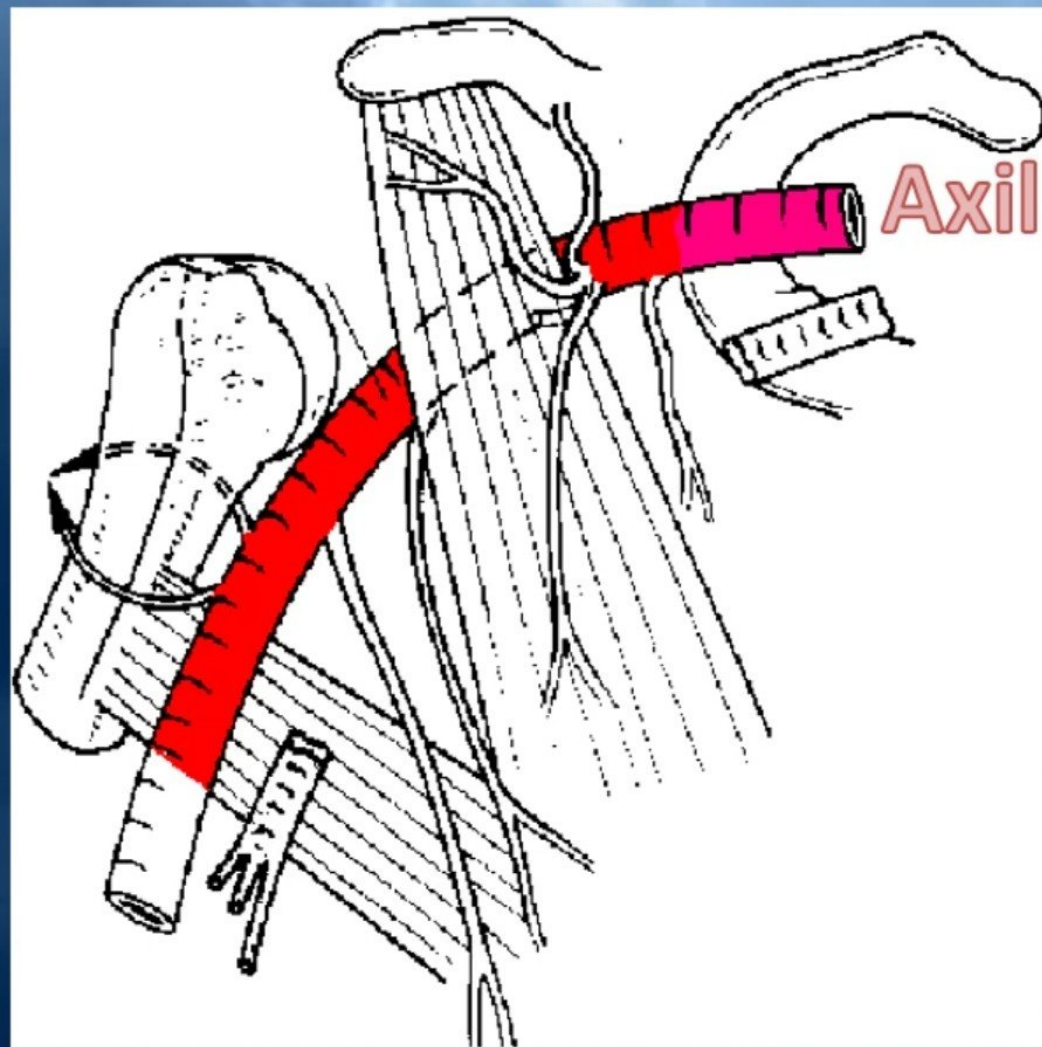


Axillary artery (Relation)

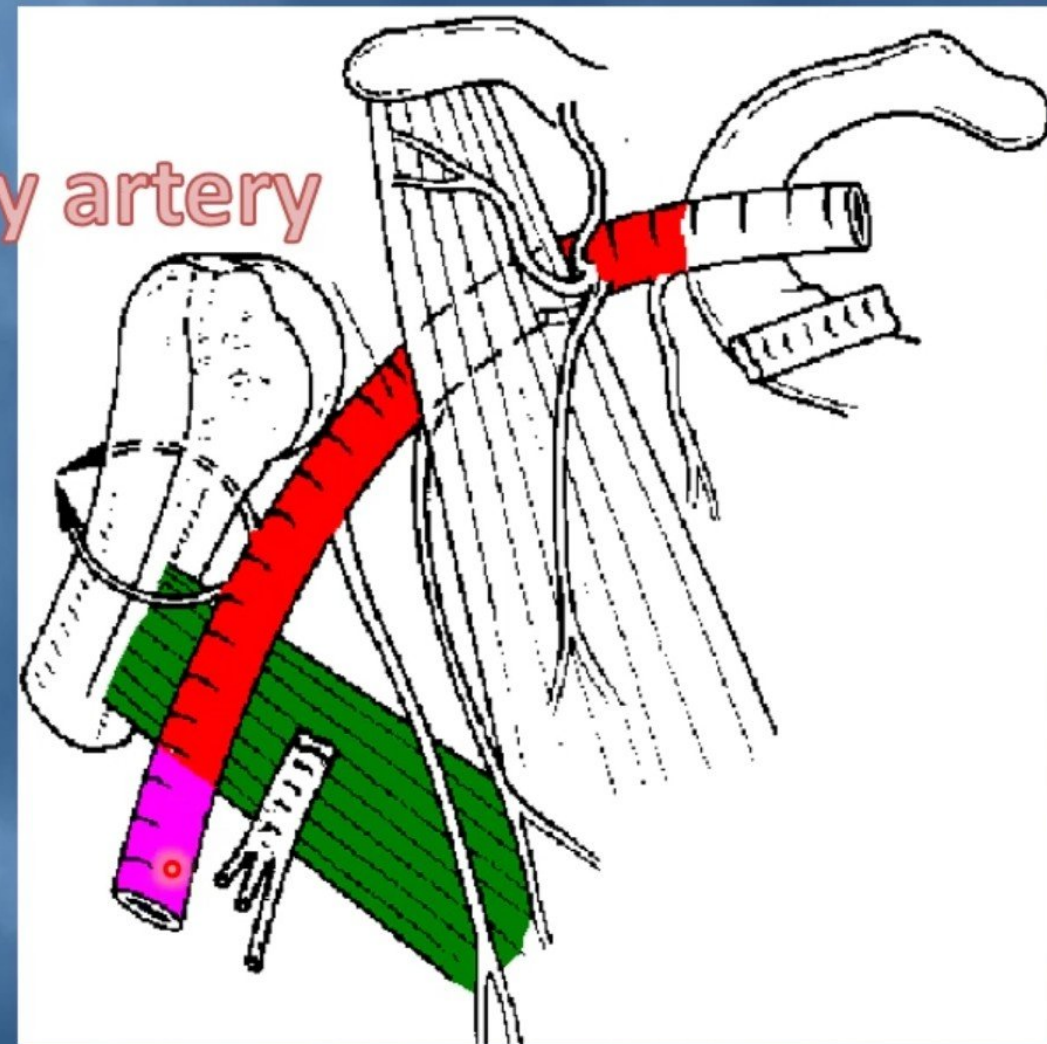




Branches of axillary artery



Axillary artery

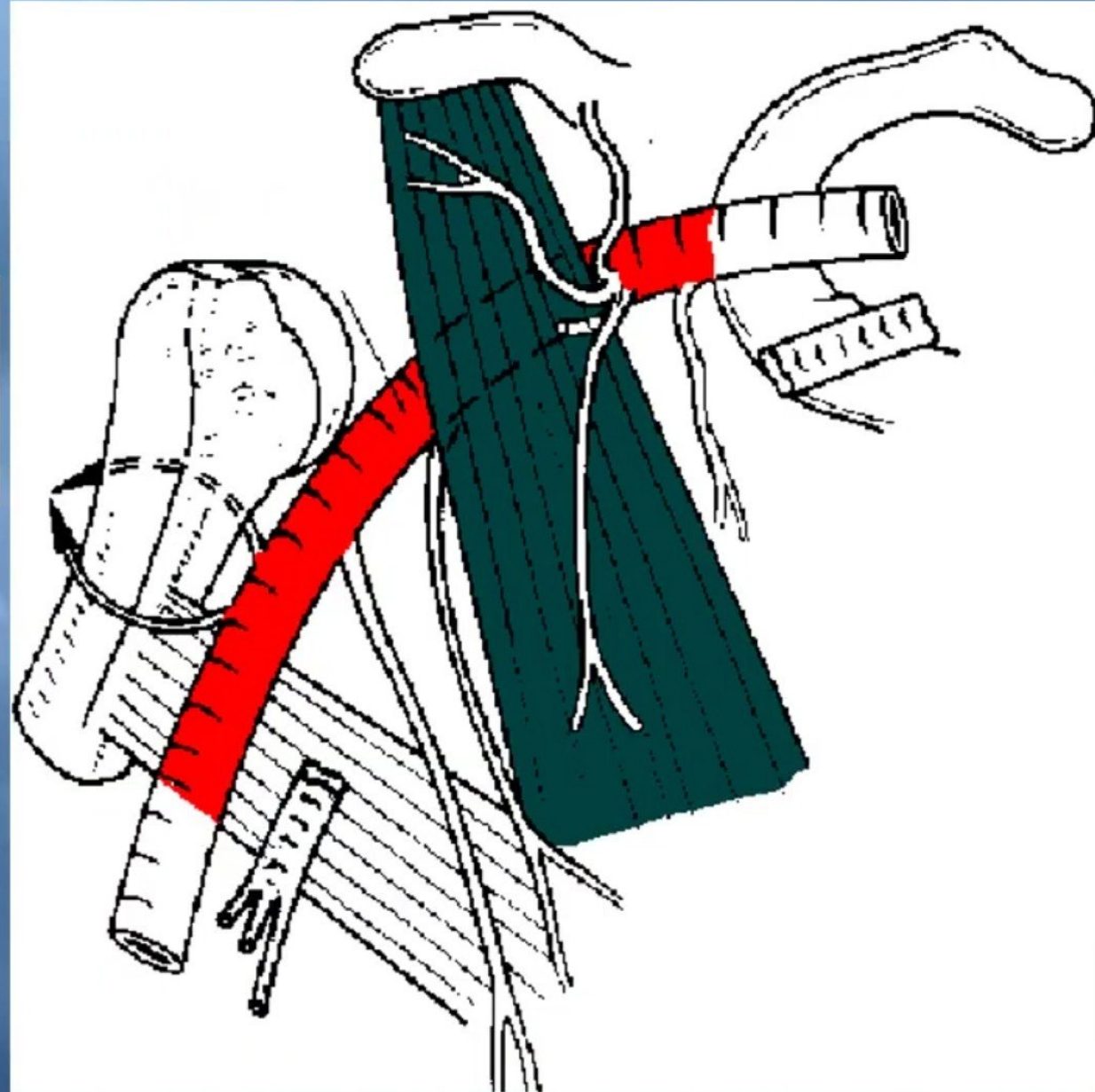


- Begins at the outer border of the first rib as the continuation of the **subclavian artery**

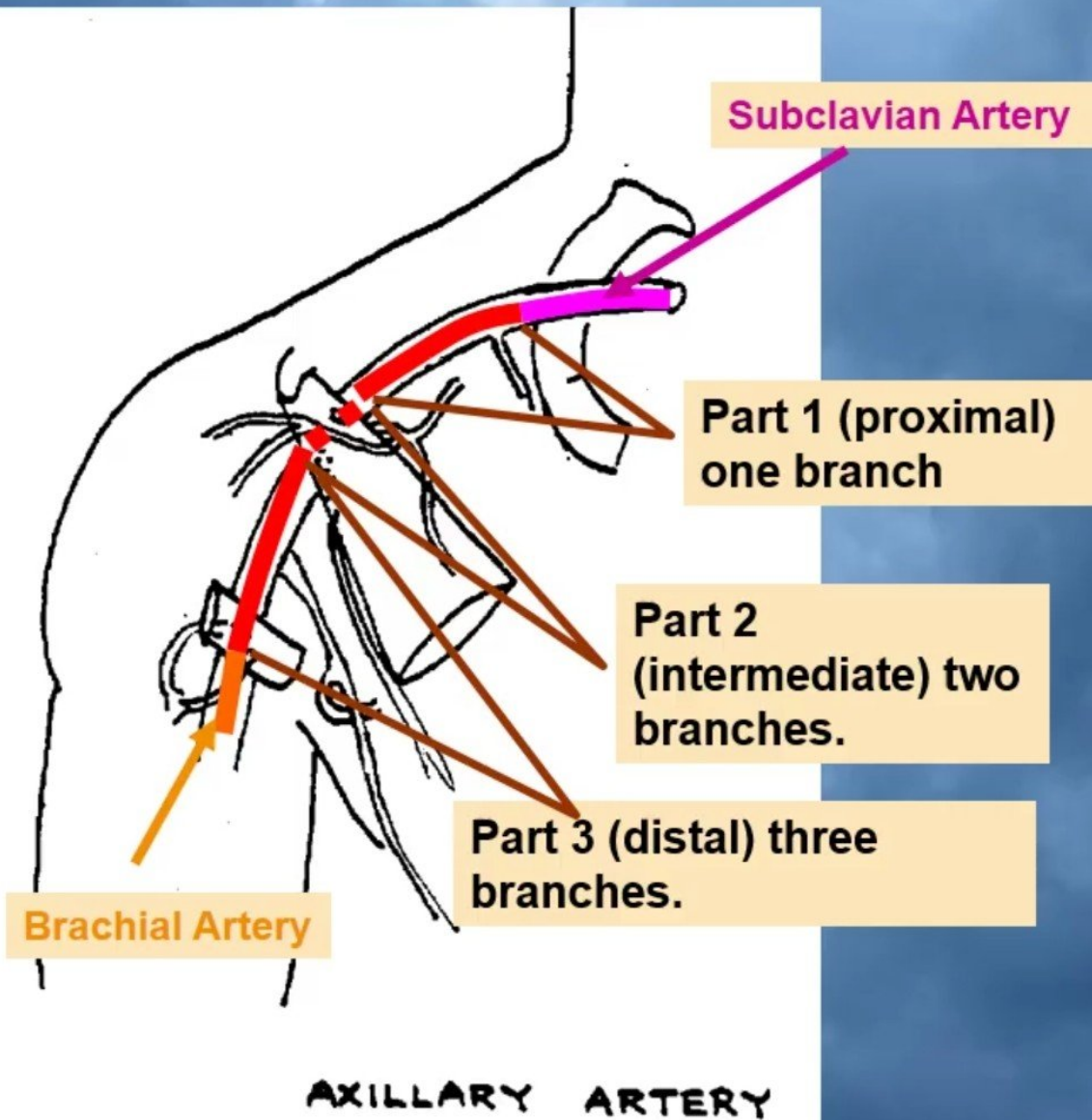
- Ends by becoming the **brachial artery** at the lower border of **teres major**

Axillary artery

- For descriptive purposes it is divided into three parts by pectoralis minor muscle

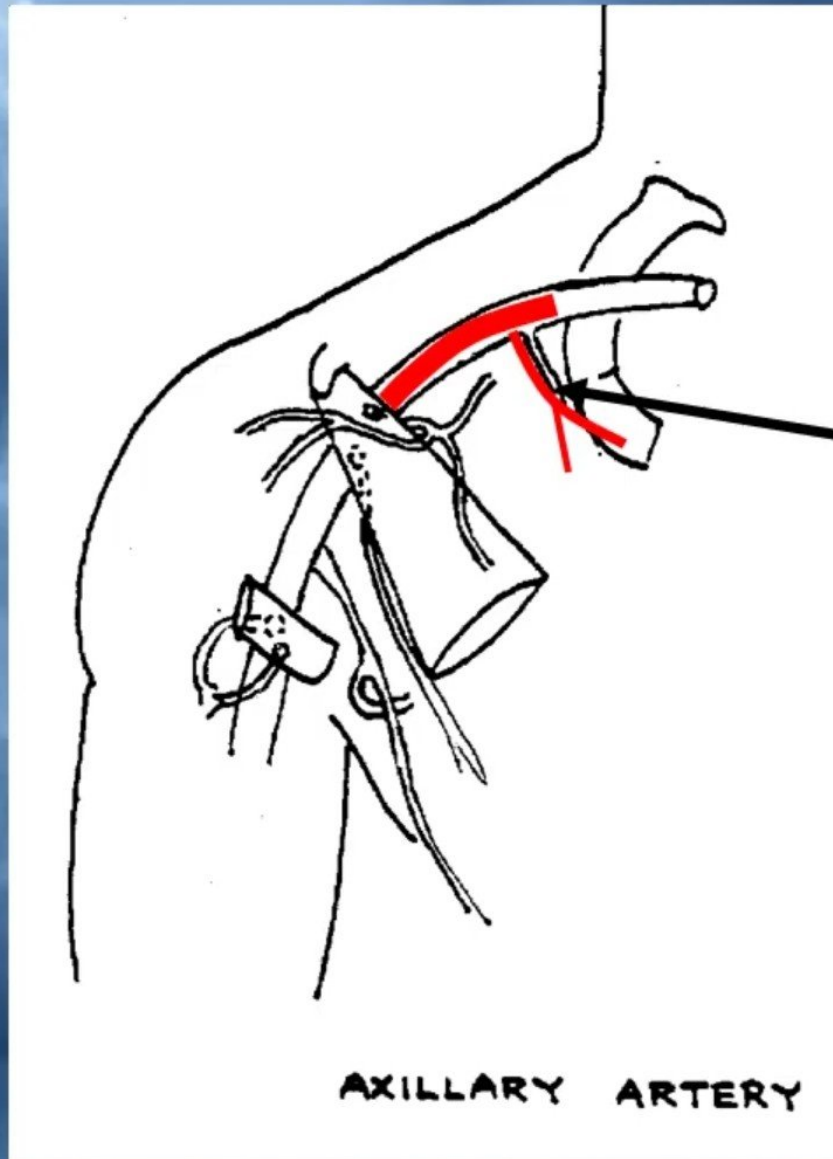


Axillary Artery: divided into three parts



Axillary Artery: First Part

From lateral border of
1st rib to medial border
of Pectoralis Major
Musles.



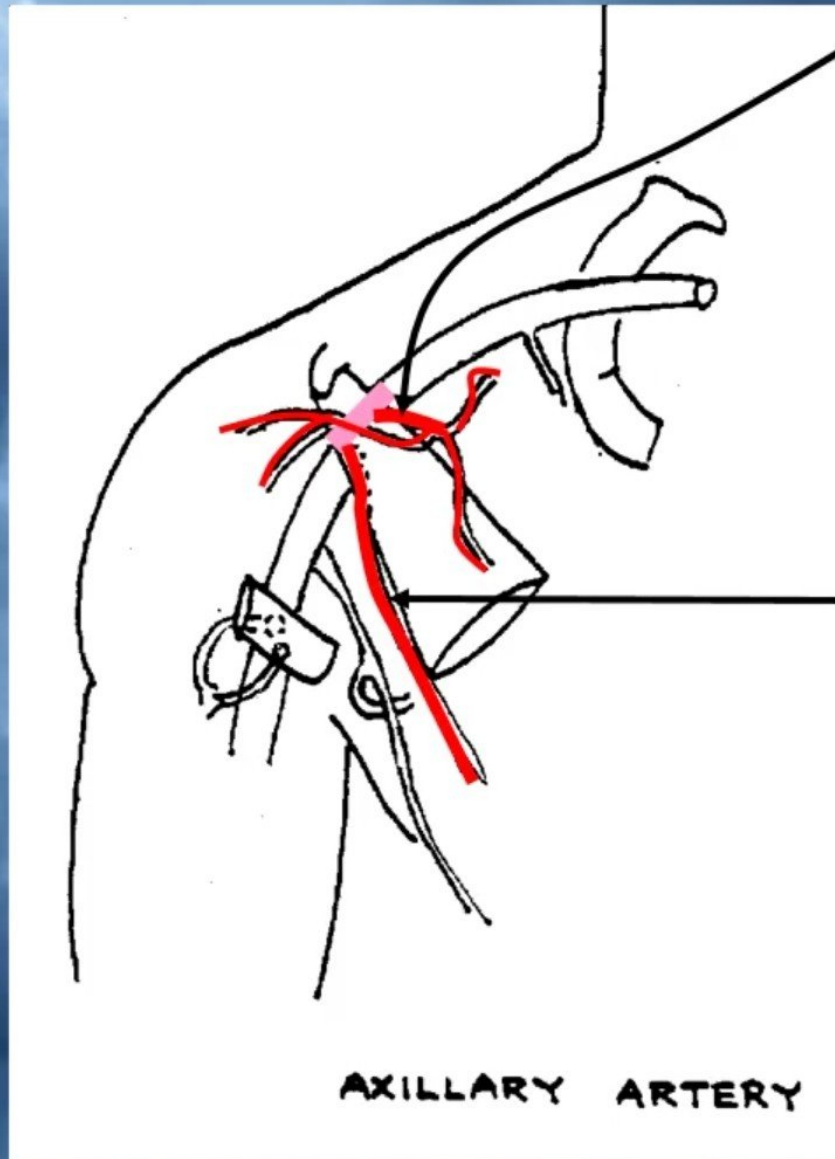
Name of Branch:

**Superior
Thoracic artery.**
(to external thoracic
body wall)

- ✓ Supplies blood
to first and
second
intercostal
spaces

Axillary Artery: **Second part**

**Deep to the
pectoralis minor
Musels.**



Thoracoacromial trunk

Branches to:

- ✓ Clavicular area.
- ✓ Pectoralis region.
- ✓ Acromion of Scapula.
- ✓ Deltoid Muscle.

Lateral Thoracic Artery

Branches to:

- ✓ Serratus Anterior Musles.

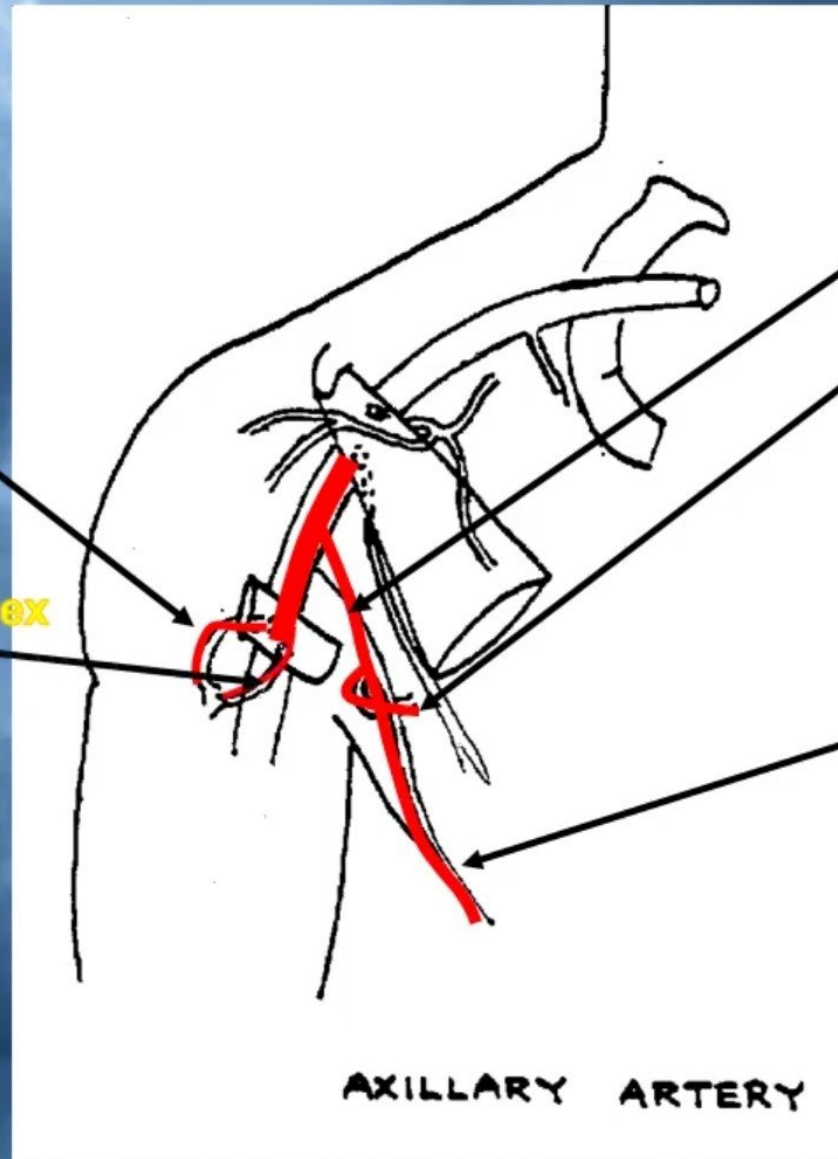
Axillary Artery:

Third part

Lateral border of Pectoralis minor Musels. to lateral border of Teres major Musels.

Posterior circumflex humeral artery.

Anterior circumflex humeral artery.



Subscapular artery:
Branches:

- 1 **Circumflex scapular artery.** (to multiple muscles associated with the scapula)
- 2 **Thoracodorsal artery.** (to Latissimus dorsi Musels)

Axillary Artery and Anastomoses Around Scapula

Anterior View

Thoracoacromial Artery.

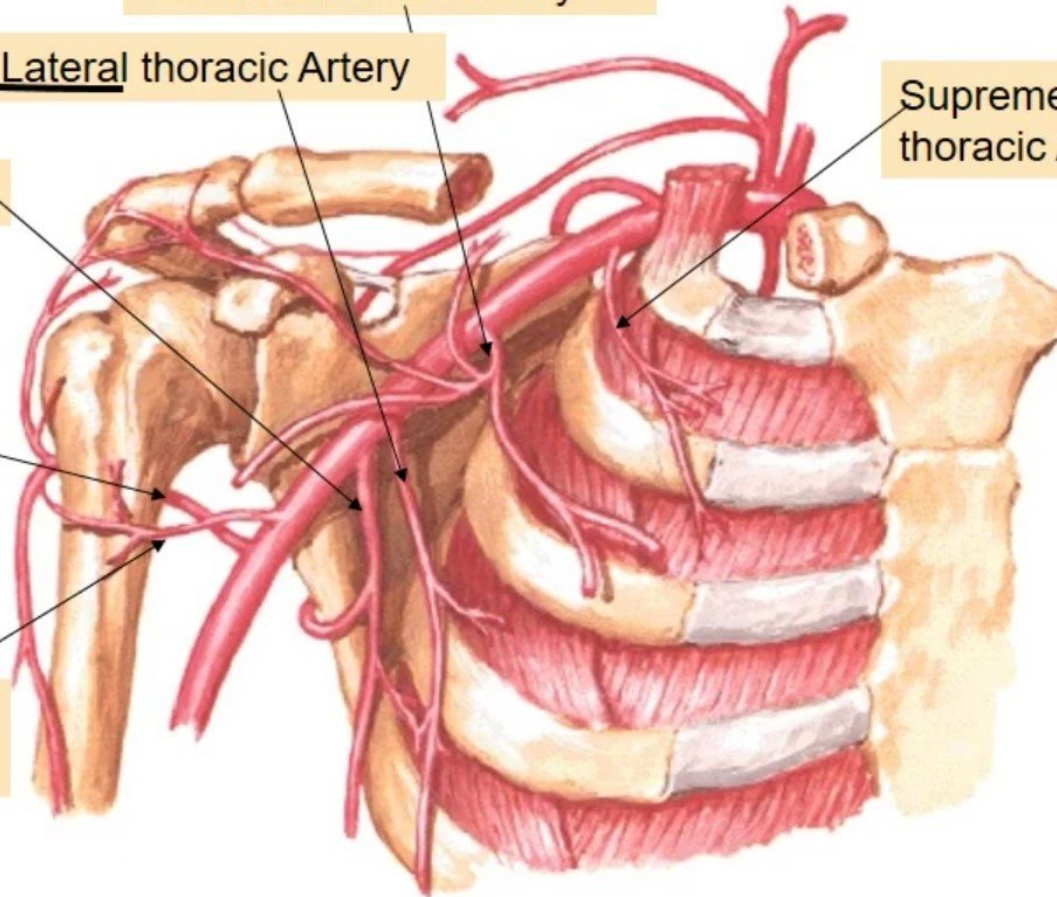
Lateral thoracic Artery

Supreme thoracic Artery.

Subscapular A.

Post. Circumflex humoral A.

Ant. Circumflex humoral A.



Second
part

First
part

Third
part

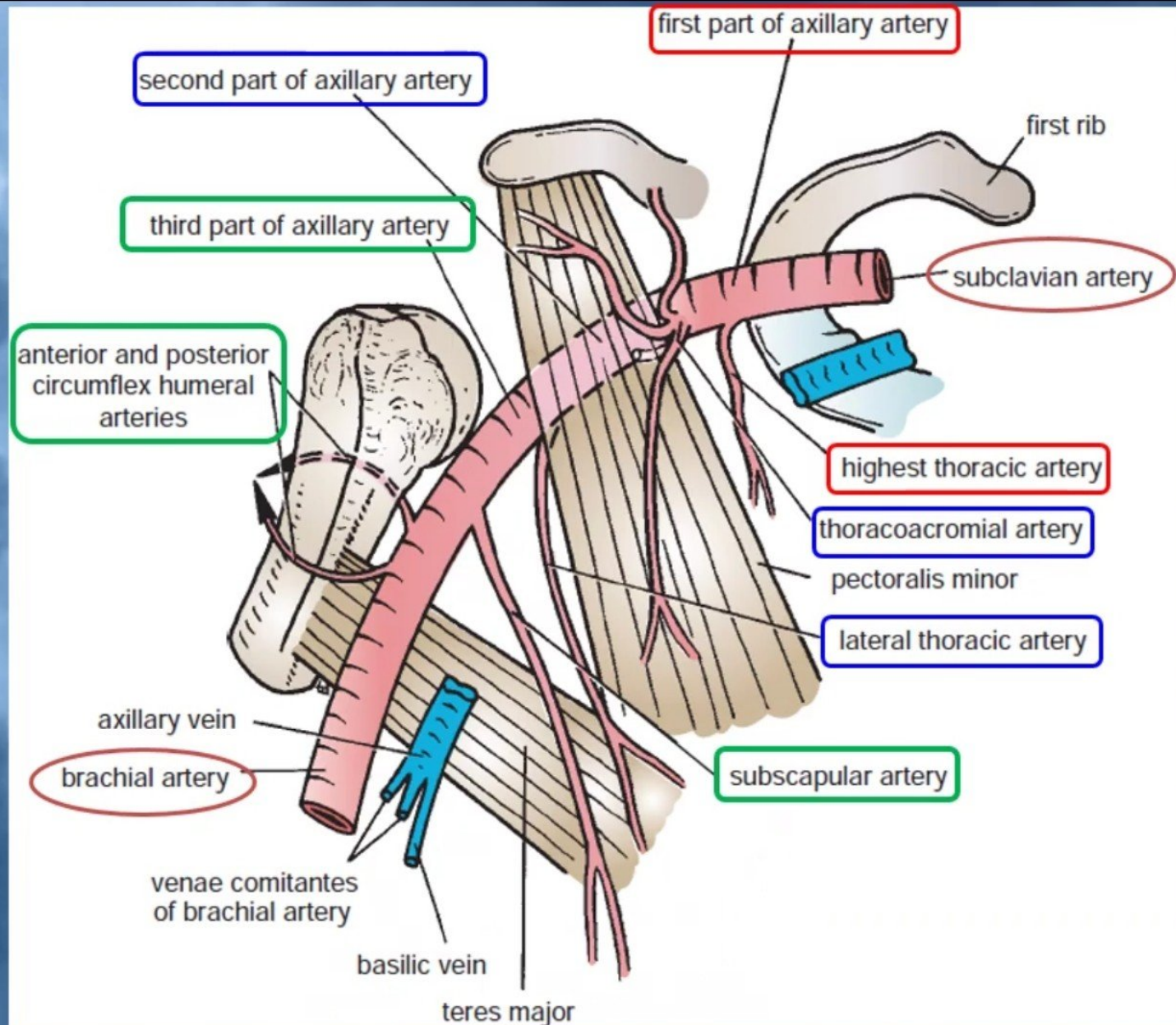
Palpation of the axillary artery

- The **third part of the axillary artery** can be palpated in the **lateral wall of the inferior part of the axilla** and can be compressed against the humerus

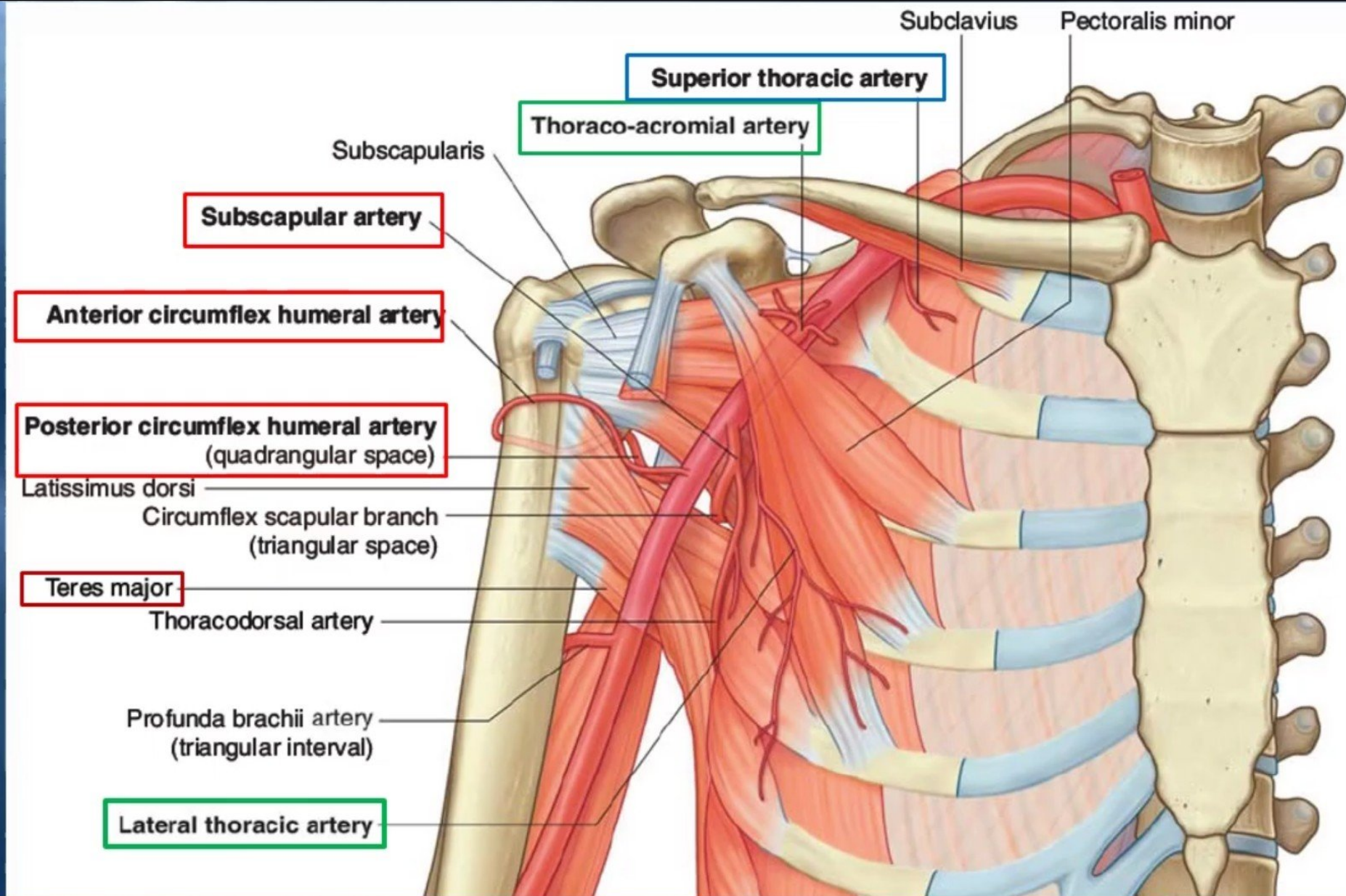


Summary for axillary artery

- ✓ Origen
- ✓ Relation
- ✓ Division
- ✓ Branches
- ✓ Termination



Summary for axillary artery



Axillary vein

Formation:

Continuation of **Basilic vein** at the lower border of **teres major**.

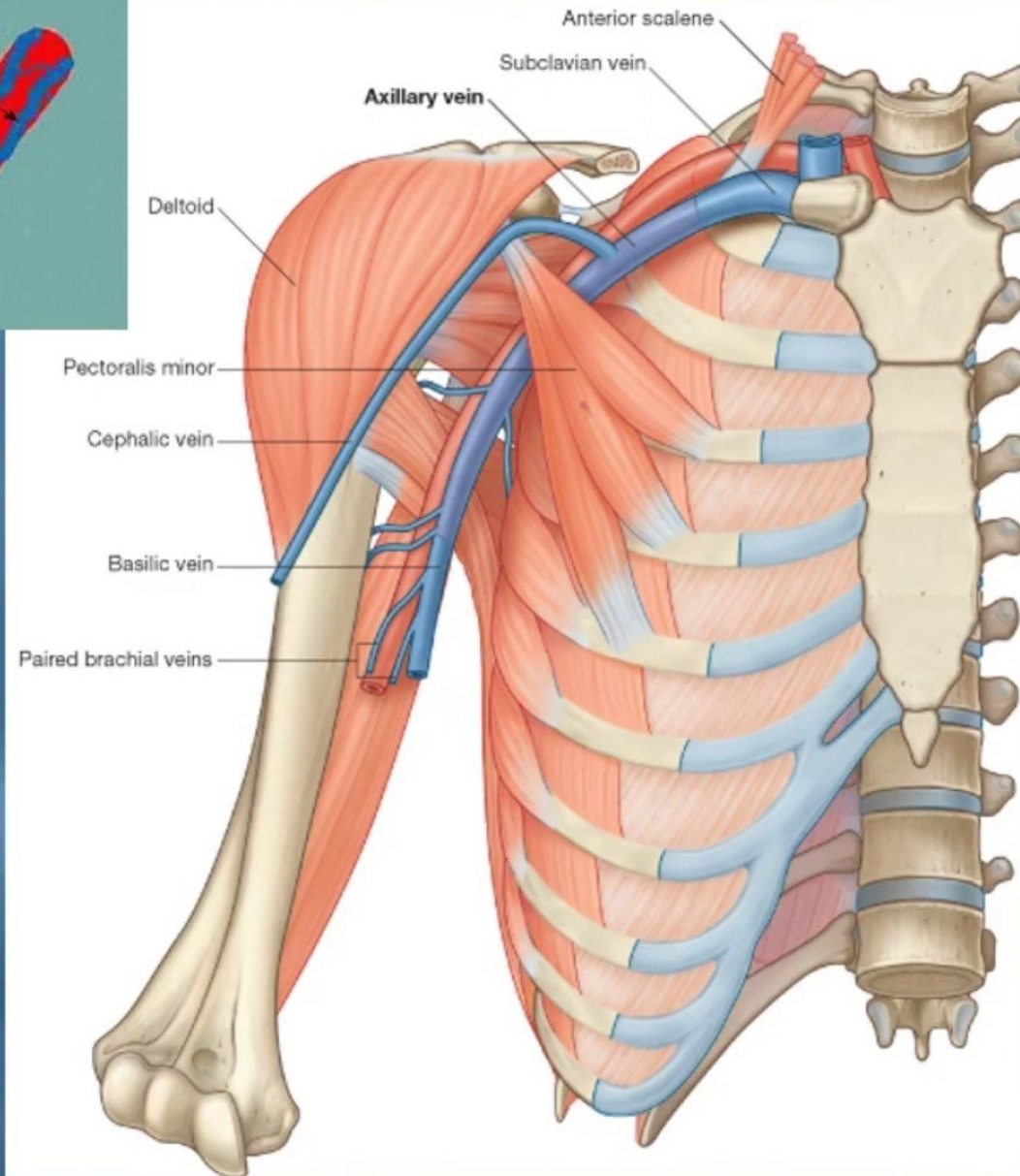
- runs upwards on the medial side of the axillary artery
- Ends at **outer border of 1st rib** by becoming the **subclavian vein**.

Tributaries:

- **Cephalic vein**.
- veins corresponding to the branches of axillary artery.

Vena comitantes

- Frequently multiple (2 or 3).
- Run with their corresponding arteries.



Axillary Lymph Nodes

1) Central Nodes

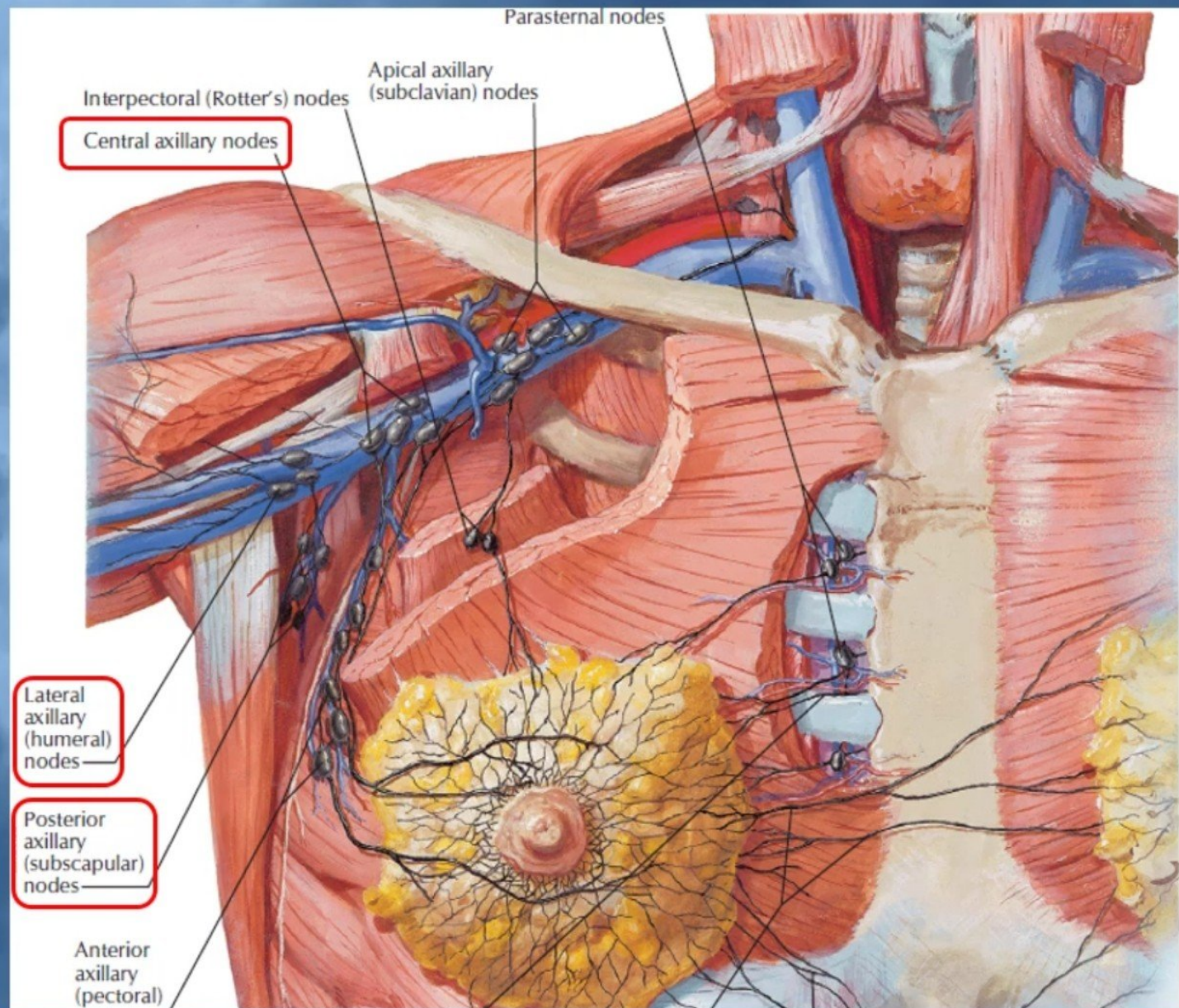
- It Lie near the base of the axilla between the lateral thoracic and subscapular veins;
- It receive lymph from the lateral, anterior, and posterior groups of nodes; and
- It drain into the apical nodes.

2) Brachial (Lateral) Nodes

- It Lie posteromedial to the axillary veins,
- It receive lymph from the upper limb, and
- It drain into the central nodes.

3) Subscapular (Posterior) Nodes

- It lie along the subscapular vein,
- It receive lymph from the posterior thoracic wall and the posterior aspect of the shoulder, and
- It drain into the central nodes.



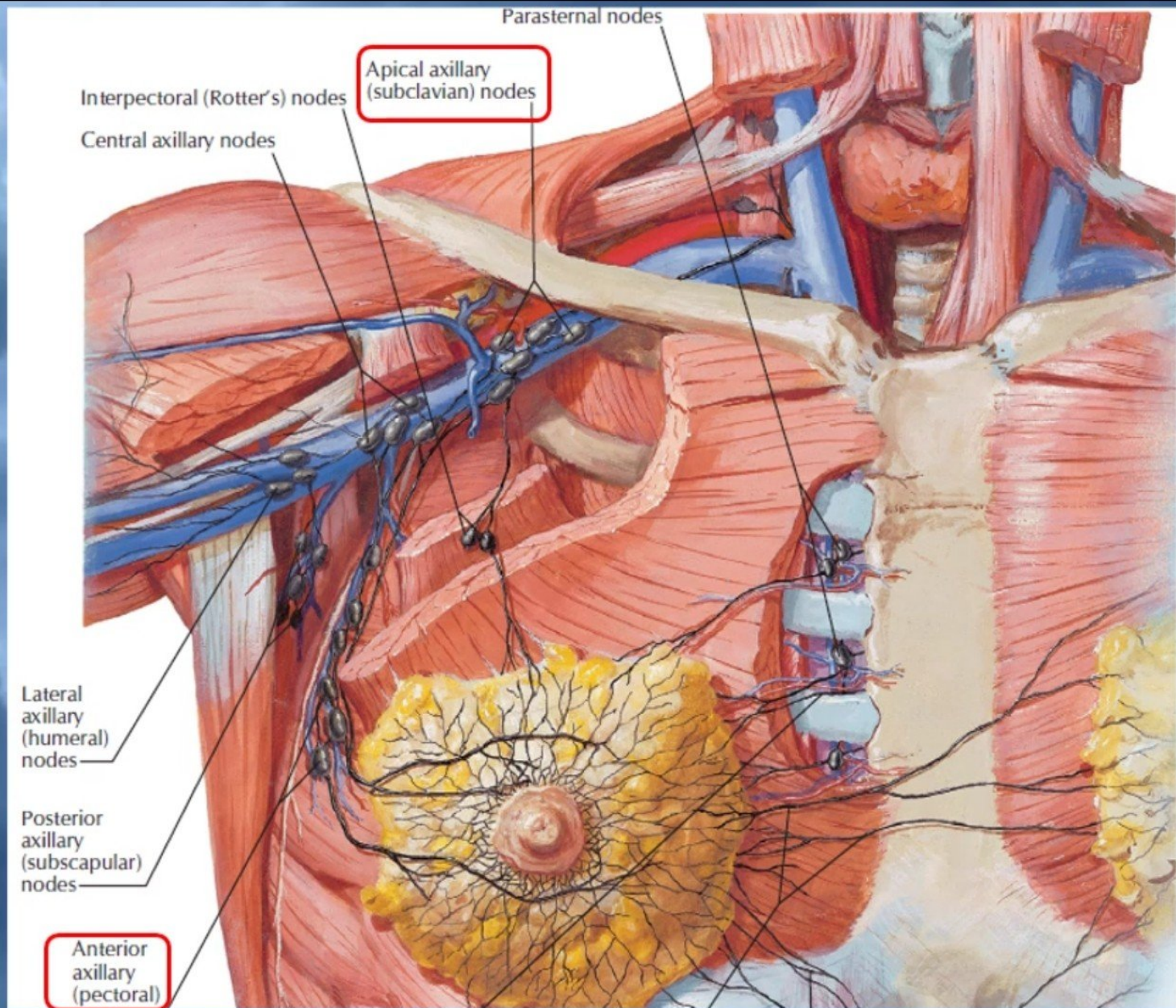
Axillary Lymph Nodes

4) Pectoral (Anterior) Nodes

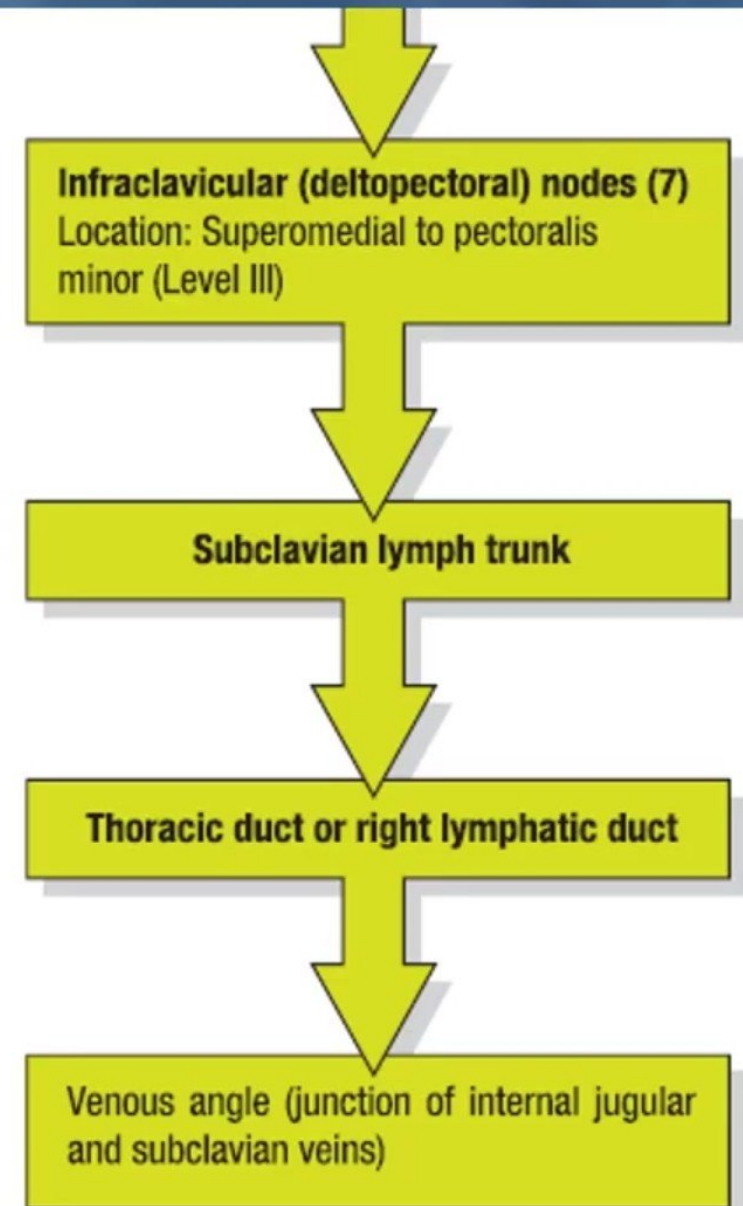
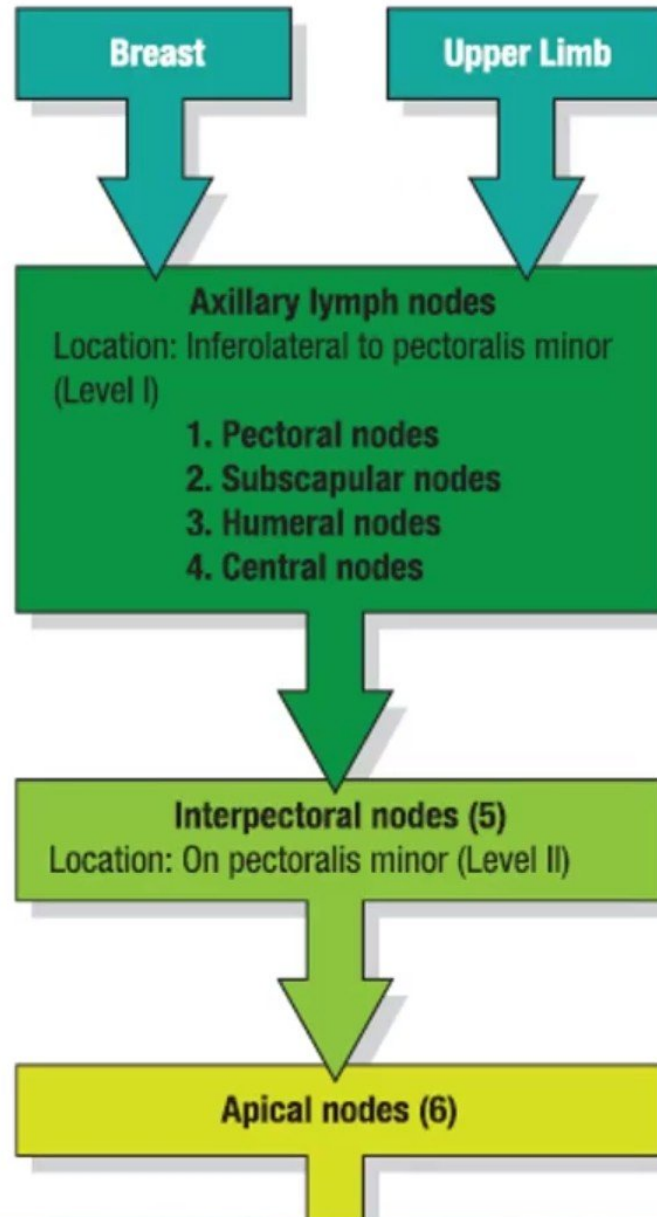
- It Lie along the inferolateral border of the pectoralis minor muscle.
- It receive lymph from the anterior and lateral thoracic walls, including the breast.
- It drain into the central nodes.

5) Apical (Medial or Subclavicular) Nodes

- It Lie at the apex of the axilla medial to the axillary vein and above the upper border of the pectoralis minor muscle,
- It receive lymph from all of the other axillary nodes (and occasionally from the breast),
- It drain into the subclavian trunks.



Follow of lymphatic derange



THANK
YOU